# Hyperspectral and Multispectral Image Fusion via Nonlocal Low-Rank Tensor Decomposition and Spectral Unmixing

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***Abstract*— Hyperspectral (HS) imaging has shown its superiority in many real applications. However, it is usually difficult to obtain high-resolution (HR) HS images through existing imaging techniques due to the hardware limitations. To improve the spatial resolution of HS images, this article proposes an effective HS-multispectral (HS-MS) image fusion method by combining the ideas of nonlocal low-rank tensor modeling and spectral unmixing. To be more precise, instead of unfolding the HS image into a matrix as done in the literature, we directly represent it as a tensor, then a designed nonlocal Tucker decomposition is used to model its underlying spatial–spectral correlation and the spatial self-similarity. The MS image serves mainly as a data constraint to maintain spatial consistency. To further reduce the spectral distortions in spatial enhancement, endmembers, and abundances from the spectral are used for spectral regularization. An efficient algorithm based on the alternating direction method of multipliers (ADMM) is developed to solve the resulting model. Extensive experiments on four HS image data sets demonstrate the superiority of the proposed method over several state-of-the-art HS-MS image fusion methods.**

***Index Terms*— Hyperspectral (HS) image, image fusion, nonlocal tensor decomposition, spatial enhancement, spectral unmixing.**

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## Nomenclature

| Notations | Explanations |
|---|---|
| $\mathcal{X}$, $\mathbf{X}$, $\mathbf{x}$, $x$ | tensor, matrix, vector, scalar. |
| $\mathbf{x}(:, i_2, i_3, \ldots, i_N)$ | fiber of tensor $\mathcal{X}$. |
| $\mathbf{X}(:, :, i_3, \ldots, i_N)$ | slice of tensor $\mathcal{X}$. |
| $\mathbf{X}_{(n)}$ or $\text{unfold}_n(\mathcal{X})$ | mode-$n$ matricization of tensor $\mathcal{X}$. |
| $\text{fold}_n(\mathbf{X}_{(n)})$ | the inverse operation of $\text{unfold}_n(\mathcal{X})$. |
| $\mathbf{Vec}(\mathcal{X})$ | vectorization of tensor $\mathcal{X}$. |
| $\mathbf{Ten}(\mathbf{x})$ | tensorization of vector $\mathbf{x}$. |
| $(r_1, r_2, \ldots, r_N)$ | multilinear rank, where $r_n = \text{Rank}(\mathbf{X}_{(n)})$. |
| $\|\mathcal{X}\|_0$ | $\ell_0$ norm of tensor $\mathcal{X}$. |
| $\|\mathcal{X}\|_1$ | $\ell_1$ norm of tensor $\mathcal{X}$. |
| $\|\mathcal{X}\|_F$ | Frobenius norm of tensor $\mathcal{X}$. |
| $\mathcal{C} = \mathcal{X} \times_n \mathbf{A}$ | mode-$n$ multiplication of $\mathcal{X}$ and $\mathbf{A}$. |

## I. Introduction

HYPERSPECTRAL (HS) imagery consists of images of one scene in tens to hundreds of discrete spectral bands at specific frequencies. The very high spectral resolution and coverage of HS images enable a precise identification of the materials present in the scene, which tend to significantly benefit the characterization of the imaged scene and greatly enhance performances in many applications including remote sensing [1]–[3], monitoring [4], object detection [5], [6], military [7], and tracking [8]. However, although achieving high spectral resolution, HS images usually have severe limitations in spatial resolution. Actually, HS imaging systems need lots of exposures to simultaneously acquire many bands within a narrow spectral window, and in order to ensure sufficient signal-to-noise ratio (SNR), long exposures are often necessary, which usually leads to the sacrifice of spatial resolution [9]. In consequence, the spatial resolution of HS images is usually poor, which limits its extensive applications to some extent due to the lots of mixed pixels in the low-resolution HS images. Therefore, enhancing the spatial resolution of HS images has become an important topic and attracted increasing interest in recent years. Simply increasing the spatial resolution of image sensors would not be effective for HS imaging because the average amount of photons reaching the sensors would be further reduced leading to even lower SNR, and thus resolution enhancement by postprocessing is a better alternative [10], [11]. On the other hand, multispectral (MS)

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imaging sensors are capable of capturing images with increasingly higher spatial resolution. Therefore, a low-resolution HS image can be fused with the high-resolution (HR) MS image captured on the same scene to reconstruct a HR HS image [12], [13]. This procedure is known as HS and MS image fusion and has attracted great attention. Actually, new HS imager has been planned which carries also an MS image sensor [14]. This means high-spectral-resolution HS image and high-spatial-resolution MS image are simultsneously captured, and more data sources for HS and MS image fusion are expected.

Pan-sharpening has been widely used to enhance the spatial resolution of MS images by the fusion of an MS image and the corresponding higher-resolution panchromatic image [15], [16]. There are mainly three categories of representative pan-sharpening techniques: component substitution (CS) [17], [18], multiresolution analysis (MRA) [19], [20], and sparse representation (SR) [16], [21]. In addition, deep learning-based methods have outperformed traditional approaches in many computer vision tasks in the last few years and have been introduced to Pan-sharpening very recently [22], [23]. These methods often suffer from serious spectral distortion because of the lack of spectral information in the panchromatic image. Obviously, pan-sharpening can be considered as a special case of the HS and MS image fusion problem, and thus those pan-sharpening methods can often be extended to the fusion of HS and MS images [24], [25]. An in-depth review of pan-sharpening methods can be found in [12].

As regards the HS and MS image fusion problem, there has been many fruitful researches in the last few years. The Bayesian framework is a frequently used methodology to fuse a low-resolution HS image with a HR MS image. This class of methods usually builds the posterior distribution of the expected HR HS images based on the prior knowledge and observation model. In [26], a Bayesian method based on a maximum a posterior (MAP) estimation and a stochastic mixing model (SMM) is proposed for the HS and MS image fusion problem. Wei *et al.* [27] solved a Sylvester equation with closed-form solution to maximize the likelihoods of the observations and meanwhile incorporate prior information for the fusion problem, which leads to a Bayesian estimator and an effective fusion algorithm with significantly decreased computational complexity. A spectral linear mixing model based on nonparametric Bayesian dictionary learning and Bayesian sparse coding is employed in [28] to solve the fusion problem.

Matrix factorization is also a commonly used methodology in HS-MS image fusion. The matrix factorization-based fusion methods usually first unfold the targeted HR HS image as a matrix, and then factorize it into a basis matrix and a coefficient matrix, where the basis and coefficient matrices are, respectively, extracted from the low-resolution HS image and HR MS image. The coupled nonnegative matrix factorization (CNMF) method proposed by Yokoya *et al.* [29] alternately unmixes the HS and MS image into endmember and abundance matrices by the CNMF algorithm based on a linear spectral mixture model. Kawakami *et al.* [30] applied an unmixing algorithm to the HS image to estimate a basis representing reflectance spectra and then solved the HR sparse coefficient using orthogonal matching pursuit. Huang *et al.* [31] proposed a sparse matrix factorization method that learns from the low-resolution HS image a spectral dictionary containing pure signatures using the $k$-singular value decomposition (SVD) algorithm. Lanaras' work [32] jointly unmixes the HS and MS images into the pure reflectance spectra and the associated mixing coefficients, which leads to a coupled matrix factorization problem with some useful constraints imposed by elementary physical properties of spectral mixing. Zhang *et al.* [33] proposed a new group spectral embedding (GSE)-based approach for HS and MS image fusion by exploring the multiple manifold structures of spectral bands and the low-rank structure of the HS image. Dong *et al.* [34] proposed a nonnegative structured SR (NSSR) approach that formulates the estimation of the HR HS image as a joint estimation of the HS dictionary and the sparse codes based on the prior knowledge of the spatial–spectral sparsity of the HS image.

Although the matrix factorization-based methods have achieved impressive performances in the HS and MS fusion problem. There are inherent drawbacks. All the matrix factorization methods need to unfold the 3-D data structures into matrices, and the unfolding procedure could break the spatial structure of the data, which makes it hard to fully exploit the spatial–spectral correlations of the HS images. Compared with low-rank matrix factorization, low-rank nonlocal tensor-based methods are able to better maintain and utilize the spatial structure of the HS images, and thus have been applied to some HS image restoration problems [35]–[37].

To maintain the inherent spatial structure of the HS images, Li *et al.* [38] proposed a coupled sparse tensor factorization (CSTF)-based approach for the HS and MS image fusion problem. Different from the conventional matrix factorization-based methods, the targeted HR HS image is directly considered as a third-order tensor which can then be estimated by a core tensor and dictionaries of the three modes. Meanwhile, a sparsity prior of the core tensor is incorporated as a regularizer to model the high spatial–spectral correlations in the HR HS image. Although achieving good performances, CSTF considers only the high spatial–spectral correlations and fails to thoroughly exploit other effective prior knowledge in the HS image. Zhang *et al.* [39] proposed a spatial–spectral-graph-regularized low-rank tensor decomposition-based fusion method. They first reformulate the fusion problem as a low-rank tensor decomposition model to utilize the low-rank property in the HS image, and then construct two graphs for the spatial consistency and spectral smoothness, respectively. These two graphs are then used as regularizers to efficiently preserve the spatial correlation and the spectral structure in the fused HR-HS images. Xu's work [40] extracts the nonlocal similar patches to form a nonlocal patch tensor and use a tensor–tensor product-based tensor SR to model the patch tensors. Then the authors built the relationship between the HS and MS images by tensor–tensor product and design a unified objective function which then is solved by alternating direction method of multipliers (ADMMs).

In the past decade, methods based on deep learning have achieved impressive performances in many computer vision tasks [41], [42] and thus have been introduced to the HS and

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MS fusion problem [43], [44]. Palsson *et al.* [43] proposed to use a 3-D convolutional neural network to fuse together HS and MS images, and meanwhile, dimensionality reduction of the HS image is performed prior to fusion in order to significantly reduce the computational time. Xie *et al.* [44] proposed a novel model-based deep learning approach to fuse HS and MS images. The authors first construct a novel fusion model and design an effective iteration algorithm to solve the model, then the designed algorithm is unfolded to a deep network with learning the proximal operators and model parameters. Although achieving good performances, deep learning-based methods usually need massive amounts of training data, which is often unpractical in the HS image restoration problems.

In this article, we proposed a nonlocal low-rank tensor decomposition and spectral unmixing based approach for dealing with the HS and MS fusion problem. To fully exploit the high spatial–spectral correlations and the nonlocal self-similarity of the HR HS image, we segment it into many overlapped 3-D patches and cluster those patches into some clusters, and then every cluster is represented as a low-rank fourth-order tensor. We use Tucker decomposition to formulate the low-rank property of those tensors. Motivated by our previous works [45], [46], we then use a 3-D total variation (3D-TV) as a regularization to formulate the spatial–spectral continuity of the HS image. Finally, a nonconvex regularization-based linear spectral unmixing is applied as a significant spectral regularization to reduce spectral distortions. Based on the above prior knowledge, we propose a unified optimization model to formulate the HS and MS fusion problem and develop an efficient algorithm using ADMM to solve the proposed model. Experimental results demonstrate that the proposed procedure substantially outperforms several state-of-the-art HS and MS fusion methods.

The rest of this article is organized as follows. In Section II, we give a basic introduction on tensors and formulate the fusion problem in tensor perspective. The proposed unified optimization model for HS and MS image fusion is then introduced in Section III. Section IV gives the ADMM-based optimization algorithm to solve the proposed model. In Section V, experimental results on four image data sets are presented. Finally, the conclusion is drawn in Section VI.

## II. Problem Formulation

Throughout this article, both HS and MS images are denoted as third-order tensors which belong to $\mathbf{R}^{w\times h\times s}$, where $w$, $h$, and $s$ are the indexes of the width, height, and spectral modes, respectively. In this section, we shall first review the basics about tensor and multilinear algebra and then formulate the fusion problem in tensor perspective.

### A. Notions and Preliminaries on Tensors

A tensor can be seen as a multiindex numerical array, and the number of its modes is called the order of this tensor. An $N$-order real-valued tensor can be denoted as $\mathcal{X} \in \mathbf{R}^{I_1\times I_2\ldots\times I_N}$ and its elements as $x_{i_1,i_2,\ldots,i_N}$, where $1\leq i_n \leq I_n$, $n = 1, 2, \ldots, N$. Then a vector can be seen as a 1-order tensor and a matrix can be seen as a two-order tensor. A subtensor is created by only using a fixed subset of the indices of the original tensor. Vector-valued subtensors are called fibers which are obtained by fixing all but one index. Matrix-valued subtensors are called slices which are defined by fixing every index but two. The $n$-mode unfolding matrix $\mathbf{X}_{(n)} =: \text{unfold}_n(\mathcal{X}) \in \mathbf{R}^{I_n\times(I_1\ldots I_{n-1}I_{n+1}\ldots I_N)}$ is defined by taking all the mode-$n$ fibers as its columns, and conversely, it can also be transformed back to the original tensor by $\mathcal{X} =: \text{fold}_n(\mathbf{X}_{(n)})$. The $n$-rank of $\mathcal{X}$, denoted as $r_n$, is defined by the rank of its $n$-mode unfolding matrix $\mathbf{X}_{(n)}$, and its multilinear rank, denoted as $(r_1, r_2, \ldots r_N)$, is the tuple of all the $n$-rank, $n = 1, 2, \ldots, N$.

The production between matrices can be generalized to the production of a tensor and a matrix. The mode-$n$ product of a tensor $\mathcal{X} \in \mathbf{R}^{I_1\times I_2\ldots\times I_N}$ by a matrix $\mathbf{A} \in \mathbf{R}^{J_n\times I_n}$, denoted by $\mathcal{X} \times_n \mathbf{A}$, is an N-order tensor $\mathcal{C} \in \mathbf{R}^{I_1\times\ldots\times J_n\times\ldots\times I_N}$, whose elements are computed by

$$c_{i_1\times\ldots\times i_{n-1}\times j_n\times i_{n+1}\ldots\times i_N} = \sum_{i_n} x_{i_1\ldots i_n\ldots i_N} a_{j_n i_n}. \tag{1}$$

The mode-$n$ production $\mathcal{C} = \mathcal{X} \times_n \mathbf{A}$ can also be calculated by the matrix multiplication $\mathbf{C}_{(n)} = \mathbf{A}\mathbf{X}_{(n)}$, then tensor $\mathcal{C}$ can be computed by $\mathcal{C} = \text{fold}_n(\mathbf{C}_{(n)})$. It is worth mentioning that for distinct modes in a series of multiplications, the order of the multiplications is irrelevant, that is

$$\mathcal{X} \times_n \mathbf{A} \times_m \mathbf{B} = \mathcal{X} \times_m \mathbf{B} \times_n \mathbf{A} (n \neq m) \tag{2}$$

and for the same modes, we have

$$\mathcal{X} \times_n \mathbf{A} \times_n \mathbf{B} = \mathcal{X} \times_n (\mathbf{B}\mathbf{A}). \tag{3}$$

The $\ell_0$ norm of $\mathcal{X}$, that is, $\|\mathcal{X}\|_0$, is defined by the number of nonzero elements of $\mathcal{X}$. The $\ell_1$ norm of $\mathcal{X}$ is defined by $\|\mathcal{X}\|_1 = \sum_{i_1,\ldots,i_N} |x_{i_1\ldots i_N}|$, and its Frobenius norm is $\|\mathcal{X}\|_F = (\sum_{i_1,\ldots,i_N} |x_{i_1\ldots i_N}|^2)^{1/2}$. For more details about tensors and their related applications (see [47]–[49]).

### B. Problem Formulation in Tensor Perspective

The fusion problem of HS and MS images is aimed at reconstructing a both high-spatial-resolution and high-spectral-resolution image from a low-spatial-resolution (LR) HS image and a low-spectral-resolution MS image from the same scene. This problem is usually considered in matrix form, that is, unfolding the third-order tensor data into matrices, which can break the inherent structures of the images. To remedy this issue, we directly tackle the HS-MS image fusion problem in tensor perspectives, which could be capable of fully exploit the spatial-spectral correlations of the HS images.

As mentioned before, both HS and MS images can be naturally represented as third-order tensors. Precisely, the HR-HS image to be reconstructed is denoted as a third-order tensor $\mathcal{X} \in \mathbf{R}^{W\times H\times S}$, where $W$, $H$, and $S$ are the dimensions of the width, height, and spectral modes, respectively. The LR-HS image is denoted as $\mathcal{Y}_h \in \mathbf{R}^{w\times h\times S}$ where $w < W$ and $h < H$, and it can be written as

$$\mathcal{Y}_h = \mathcal{X} \times_1 \mathbf{P_1} \times_2 \mathbf{P_2} \tag{4}$$

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where $\mathbf{P_1} \in \mathbf{R}^{w\times W}$ and $\mathbf{P_2} \in \mathbf{R}^{h\times H}$ are the downsampling matrices along the width mode and height mode, respectively, which describe the spatial response of the imaging sensors. Specifically, (4) is actually a formal description model, and in our method the downsampling matrices, respectively, along the width mode and height mode, $\mathbf{P_1}$ and $\mathbf{P_2}$, are actually not needed. In the viewpoint of matricization, $\mathcal{X}$ and $\mathcal{Y}_h$ are usually unfolded as $\mathbf{X}_{(3)} \in \mathbf{R}^{S\times(W\times H)}$ and $(\mathbf{Y}_h)_{(3)} \in \mathbf{R}^{S\times(w\times h)}$, and then $(\mathbf{Y}_h)_{(3)}$ can be written as

$$(\mathbf{Y}_h)_{(3)} = \mathbf{X}_{(3)}\mathbf{M} \tag{5}$$

where $\mathbf{M} \in \mathbf{R}^{(W\times H)\times(w\times h)}$ denotes the degradation operator of blurring and downsampling associated with the LR-HS image. Equation (4) means that the point spread function (PSF) of the HS sensor and the downsampling matrices of the width mode and height modes are separable, then the action of the spatial subsampling matrix $M$ is decoupled with respect to the two spatial modes of $\mathcal{X}$

$$\mathbf{M} = (\mathbf{P_2} \otimes \mathbf{P_1})^T. \tag{6}$$

Separability assumption is sometimes necessary, e.g., in [38] both $\mathbf{P_1}$ and $\mathbf{P_2}$ are needed in their method. As for our method, such assumption is not necessary because only $\mathbf{M}$ is needed which is consistent with most of the existing matrix-based methods.

The HR-MS image is denoted as $\mathcal{Y}_m \in \mathbf{R}^{W\times H\times s}$ where $s < S$ is the number of bands in the HR-MS image, and it can be written as

$$\mathcal{Y}_m = \mathcal{X} \times_3 \mathbf{P_3} \tag{7}$$

where $P_3 \in \mathbf{R}^{s\times S}$ is the downsampling matrix of the spectral mode.

Reconstructing $\mathcal{X}$ from $\mathcal{Y}_h$ and $\mathcal{Y}_m$ is obviously an inverse problem, and to get satisfactory $\mathcal{X}$, it is necessary to fully exploit the prior information of $\mathcal{X}$. In the following section, we will fully analyze the useful priors and characterize them using tensor algebra before describing our final model.

## III. Proposed Model for HS-MS Fusion

With the above discussions, we should reconstruct $\mathcal{X}$ based on

$$\mathcal{Y}_h = \mathcal{X} \times_1 \mathbf{P_1} \times_2 \mathbf{P_2}, \mathcal{Y}_m = \mathcal{X} \times_3 \mathbf{P_3} \tag{8}$$

where the spatial blurring and downsampling matrices, $\mathbf{P_1}$, $\mathbf{P_2}$, or their coupled form $\mathbf{M} = (\mathbf{P_2}\otimes\mathbf{P_1})^T$, and the spectral downsampling matrices **P3**, are assumed to be known. A nature procedure is to solve the following model:

$$\min_{\mathcal{X}} \frac{1}{2}\|\mathcal{Y}_h - \mathcal{X} \times_1 \mathbf{P_1} \times_2 \mathbf{P_2}\|_F^2 + \frac{\lambda_1}{2}\|\mathcal{Y}_m - \mathcal{X} \times_3 \mathbf{P_3}\|_F^2. \tag{9}$$

However, solving (9) cannot get satisfactory result, because it is too simplified and has not considered any prior information of $\mathcal{X}$ at all. In the following, we shall analyze the inherent structures of the HR-HS image $\mathcal{X}$ in detail and depict mathematically the regularization for the model in (9).

### A. Underlying Priors of $\mathcal{X}$

It is known that $\mathcal{X}$ shows strong spectral correlation which implies that its matrix unfolding in the spectral mode $\mathbf{X}_{(3)}$ should be low rank. Meanwhile, we can consider its nonlocal self-similarity to further exploit the spatial correlation. Spatial nonlocal self-similarity is a patch-based powerful prior and has been widely used in the community of image processing [50], [51]. This prior means that one local patch in one image has many similar structure patches, and the similarity of patches actually implies the correlation of those patches. Inspired by the work of Cao *et al.* [52], we can segment $\mathcal{X}$ into many overlapped 3-D patches of the size $d \times d \times S$ and then collect these 3-D patches as a patch set $\mathcal{G}$: $\mathcal{G} = \{\mathcal{P}_i \in \mathbf{R}^{d\times d\times S} : i \in \Gamma\}$, where $\Gamma$ indicates the index set and $\mathcal{P}_i$ is the $i$th 3-D patch in the set. We cluster the patch set $\mathcal{G}$ into $K$ clusters and then collect each cluster as a fourth-order tensor, denoted by $\mathcal{Q}_p(\mathcal{X}) \in \mathbf{R}^{d\times d\times S\times N_p}$, $p = 1, 2, \ldots, K$, where $N_p$ is the number of 3-D patches in every cluster. Due to the similarity of 3-D patches in every cluster, for every $\mathcal{Q}_p(\mathcal{X})$, its matrix unfolding in the patch mode $\mathcal{Q}_p(\mathcal{X})_{(4)}$ should also be low rank.

Tucker decomposition [53] is a well-known form of tensor decomposition and has shown its advantages in many applications. Every low rank tensor $\mathcal{Q}_p(\mathcal{X})$ can be approximated by the following Tucker decomposition form:

$$\mathcal{Q}_p(\mathcal{X}) = \mathcal{C}_p \times_1 \mathbf{W}_p \times_2 \mathbf{H}_p \times_3 \mathbf{S}_p \times_4 \mathbf{N}_p + \mathcal{E}_p \tag{10}$$

where factor matrices $\mathbf{W}_p$, $\mathbf{H}_p$, $\mathbf{S}_p$, and $\mathbf{N}_p$ are orthogonal in columns for height, width, spectral, and patch modes, respectively, that is, $\mathbf{W}_p^T\mathbf{W}_p = \mathbf{I}, \mathbf{H}_p^T\mathbf{H}_p = \mathbf{I}, \mathbf{S}_p^T\mathbf{S}_p = \mathbf{I}, \mathbf{N}_p^T\mathbf{N}_p = \mathbf{I}$, to avoid the indeterminacy of scales of $\mathbf{W}_p$, $\mathbf{H}_p$, $\mathbf{S}_p$, and $\mathbf{N}_p$, $\mathcal{C}_p \in \mathbf{R}^{r_1\times r_2\times r_3\times r_4}$ is the core tensor interacts with these factors, and $\mathcal{E}_p$ is the disturbance. Considering the low-rank property in spectral and patch modes of $\mathcal{Q}_p(\mathcal{X})$, $r_3$, and $r_4$ should satisfy $r_3 \ll S$, $r_4 < N$. After that all the fourth-order tensor $\mathcal{Q}_p(\mathcal{X})$ has been approximatively estimated by the low rank tensor $\mathcal{L}_p$

$$\mathcal{L}_p = \mathcal{C}_p \times_1 \mathbf{W}_p \times_2 \mathbf{H}_p \times_3 \mathbf{S}_p \times_4 \mathbf{N}_p. \tag{11}$$

$\mathcal{X}$ can then be reconstructed by arranging those 3-D patches in the original order, that is, $\mathcal{X} = \mathcal{Q}^{-1}(\{\mathcal{L}_p\}_{p=1,2,\ldots,K})$, where $\{\mathcal{L}_p\}_{p=1,2,\ldots,K}$ is the set of $\mathcal{L}_p$ and $\mathcal{Q}^{-1}$ is the reconstruction operator.

Besides the spatial–spectral correlation and nonlocal self-similarity, $\mathcal{X}$ also shows spatial–spectral continuity which can be formulated by total variation (TV) regularization. TV regularization has been -widely used to explore the spatial piecewise smooth structure for tackling the HS image restoration task [54]–[56]. Considering that there is also a strong local smooth structure of a HS image along its spectral mode, we thus use a 3D-TV to model the spatial–spectral continuity of $\mathcal{X}$. The 3D-TV of $\mathcal{X}$ can be defined by the following form:

$$\begin{aligned}\|\mathcal{X}\|_{\text{3D-TV}} := \sum_{i,j,k}\{&|\mathcal{X}(i,j,k) - \mathcal{X}(i+1,j,k)| \\ &+ |\mathcal{X}(i,j,k) - \mathcal{X}(i,j+1,k)| \\ &+ |\mathcal{X}(i,j,k) - \mathcal{X}(i,j,k+1)|\}\end{aligned} \tag{12}$$

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which can be rewritten as the following equivalent form:

$$\|\mathcal{X}\|_{\text{3D-TV}} = \|\mathbf{D}(\mathcal{X})\|_1 \tag{13}$$

where $\mathbf{D}(\cdot) = [\mathbf{D}_w(\cdot), \mathbf{D}_h(\cdot), \mathbf{D}_s(\cdot)]$ is the 3-D difference operator, and $\mathbf{D}_w(\cdot), \mathbf{D}_h(\cdot), \mathbf{D}_s(\cdot)$ are the first-order difference operators along the width, height, and spectral modes, respectively.

With the above discussions, the model (9) is regularized as

$$\begin{aligned}\min_{\substack{\mathcal{X},\mathcal{C}_p,\mathbf{W}_p\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{V}_p}} & \frac{1}{2}\|\mathcal{Y}_h - \mathcal{X}\times_1 \mathbf{P_1}\times_2 \mathbf{P_2}\|_F^2 + \frac{\lambda_1}{2}\|\mathcal{Y}_m - \mathcal{X}\times_3 \mathbf{P_3}\|_F^2 \\ & + \lambda_2\|\mathcal{X}\|_{\text{3D-TV}} \\ \text{s.t. } & \mathcal{X} = \mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1 \mathbf{W}_p\times_2 \mathbf{H}_p\times_3 \mathbf{S}_p\times_4 \mathbf{N}_p\}_{p=1,2,\ldots,K}) \\ & \mathbf{W}_p^T\mathbf{W}_p = \mathbf{I},\quad \mathbf{H}_p^T\mathbf{H}_p = \mathbf{I},\ \mathbf{S}_p^T\mathbf{S}_p = \mathbf{I},\ \mathbf{N}_p^T\mathbf{N}_p = \mathbf{I}.\end{aligned} \tag{14}$$

### *B. Linear Spectral Unmixing Based on Minmax Concave Plus (MCP) Regularization*

The above model (14) only considers resolution enhancement, and such processing scheme tends to suffer from spectral distortions. Spectral unmixing [57], [58] has been used as a significant spectral regularization to reduce spectral distortions [59]. Once unfolding the HR-HS image $\mathcal{X}$ as $\mathbf{X}_{(3)} \in \mathbf{R}^{S\times(W\times H)}$, then a sparse spectral unmixing model is formulated as

$$\min_{\mathbf{E},\mathbf{U}} \frac{1}{2}\|\mathbf{X}_{(3)} - \mathbf{E}\mathbf{U}\|_F^2 + \lambda\|\mathbf{U}\|_0,\quad \text{s.t. } \mathbf{E}\geq 0,\ \mathbf{U}\geq 0 \tag{15}$$

where $\mathbf{E} \in \mathbf{R}^{S\times J}$ contains $J$ endmembers, while $\mathbf{U} \in \mathbf{R}^{J\times(W\times H)}$ represents fraction abundances and is regularized using sparsity constraint $\ell_0$ norm. Due to the nonconvexity of $\ell_0$ norm, the sparsity constraint is often replaced by its convex relaxation form, usually $\ell_1$ norm

$$\min_{\mathbf{E},\mathbf{U}} \frac{1}{2}\|\mathbf{X}_{(3)} - \mathbf{E}\mathbf{U}\|_F^2 + \lambda\|\mathbf{U}\|_1,\quad \text{s.t. } \mathbf{E}\geq 0,\ \mathbf{U}\geq 0. \tag{16}$$

$\ell_1$ norm is a frequently used sparsity constraint, however, several studies [60], [61] have shown that the $\ell_1$ norm (or LASSO) penalty over-penalizes large entries of vectors, and usually cannot avoid modeling bias. Fortunately, the folded-concave penalization such as smoothly clipped absolute deviation (SCAD) and MCP have been proposed and they are being describing as nearly unbiased and superior to the convex penalization approach through numerous theoretical and numerical studies [61]–[64]. In this article, we use MCP rather than $\ell_1$ norm as the sparsity constraint to achieve better unmixing performance.

MCP is a typical folded-concave penalty function, and its form is as follows:

$$\psi_{a,\lambda}(t) := \begin{cases} \dfrac{a\lambda^2}{2}, & \text{if } |t|\geq a\lambda \\ \lambda|t| - \dfrac{t^2}{2a}, & \text{otherwise.}\end{cases} \tag{17}$$

Its derivative is given by

$$\psi'_{a,\lambda}(t) = \left(\lambda - \frac{t}{a}\right)_+,\quad \text{for some } a>1 \tag{18}$$

where $(x)_+ = \max(x, 0)$. Using MCP penalty as the sparsity constraint, the above sparse unmixing model is formulated as

$$\min_{\mathbf{E},\mathbf{U}} \frac{1}{2}\|\mathbf{X}_{(3)} - \mathbf{E}\mathbf{U}\|_F^2 + \lambda\|\mathbf{U}\|_{\text{MCP}},\quad \text{s.t. } \mathbf{E}\geq 0,\ \mathbf{U}\geq 0. \tag{19}$$

Except for the different spatial resolution, both the HR-HS image $\mathcal{X}$ and LR-HS image $\mathcal{Y}_h$ actually capture the same scene, and thus their endmembers should also be the same [65], [66]. Based on this, we can simultaneously unmix $\mathbf{X}_{(3)}$ and $(\mathbf{Y}_h)_{(3)}$, then they should have the same endmembers $\mathbf{E}$ and different abundances $\mathbf{U_1}$ and $\mathbf{U_2}$. We can use this sparse unmixing approach as a significant spectral regularization in model (14) to reduce spectral distortions, which leads to our final model:

$$\begin{aligned}\min_{\substack{\mathcal{X},\mathcal{C}_p,\mathbf{W}_p\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{V}_p}} & \frac{1}{2}\|\mathcal{Y}_h - \mathcal{X}\times_1 \mathbf{P_1}\times_2 \mathbf{P_2}\|_F^2 + \frac{\lambda_1}{2}\|\mathcal{Y}_m - \mathcal{X}\times_3 \mathbf{P_3}\|_F^2 \\ & + \lambda_2\|\mathcal{X}\|_{\text{3D-TV}} + \frac{\lambda_3}{2}\|\mathbf{X}_{(3)} - \mathbf{E}\mathbf{U_1}\|_F^2 + \lambda_4\|\mathbf{U_1}\|_{\text{MCP}} \\ & + \frac{\lambda_5}{2}\|(\mathbf{Y}_h)_{(3)} - \mathbf{E}\mathbf{U_2}\|_F^2 + \lambda_6\|\mathbf{U_2}\|_{\text{MCP}} \\ \text{s.t. } & \mathcal{X} = \mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1 \mathbf{W}_p\times_2 \mathbf{H}_p\times_3 \mathbf{S}_p\times_4 \mathbf{N}_p\}_{p=1,2,\ldots,K}) \\ & \mathbf{W}_p^T\mathbf{W}_p = \mathbf{I},\quad \mathbf{H}_p^T\mathbf{H}_p = \mathbf{I},\ \mathbf{S}_p^T\mathbf{S}_p = \mathbf{I}\mathbf{N}_p^T\mathbf{N}_p = \mathbf{I} \\ & \mathbf{E}\geq 0,\quad \mathbf{U_1}\geq 0,\ \mathbf{U_2}\geq 0.\end{aligned} \tag{20}$$

## IV. Optimization Procedure

### *A. ADMM-Based Solving Algorithm*

We now optimize the model (20) using a multiblock version of ADMM [67]–[69]. Mathematically, (20) has the following equivalent form:

$$\begin{aligned}\min_{\substack{\mathcal{A},\mathcal{B},\mathcal{F},\mathcal{Z},\\ \mathcal{X},\mathcal{C}_p,\mathbf{W}_p,\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{V}_p}} & \frac{1}{2}\|\mathcal{Y}_h - \mathcal{A}\times_1 \mathbf{P_1}\times_2 \mathbf{P_2}\|_F^2 + \frac{\lambda_1}{2}\|\mathcal{Y}_m - \mathcal{B}\times_3 \mathbf{P_3}\|_F^2 \\ & + \lambda_2\|\mathcal{F}\|_1 + \frac{\lambda_3}{2}\|\mathbf{G}_{(3)} - \mathbf{E}\mathbf{U_1}\|_F^2 + \lambda_4\|\mathbf{V_1}\|_{\text{MCP}} \\ & + \frac{\lambda_5}{2}\|(\mathbf{Y}_h)_{(3)} - \mathbf{E}\mathbf{U_2}\|_F^2 + \lambda_6\|\mathbf{V_2}\|_{\text{MCP}} \\ \text{s.t. } & \mathcal{X} = \mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1 \mathbf{W}_p\times_2 \mathbf{H}_p\times_3 \mathbf{S}_p\times_4 \mathbf{N}_p\}_{p=1,2,\ldots,K}) \\ & \mathbf{W}_p^T\mathbf{W}_p = \mathbf{I},\quad \mathbf{H}_p^T\mathbf{H}_p = \mathbf{I},\ \mathbf{S}_p^T\mathbf{S}_p = \mathbf{I},\ \mathbf{N}_p^T\mathbf{N}_p = \mathbf{I} \\ & \mathbf{E}\geq 0,\quad \mathbf{V_1}\geq 0,\ \mathbf{V_2}\geq 0 \\ & \mathcal{A} = \mathcal{X},\quad \mathcal{B} = \mathcal{X},\ \mathcal{F} = \mathbf{D}(\mathcal{Z}) \\ & \mathcal{Z} = \mathcal{X},\quad \mathcal{G} = \mathcal{X},\ \mathbf{U_1} = \mathbf{V_1},\ \mathbf{U_2} = \mathbf{V_2}\end{aligned} \tag{21}$$

where $\mathbf{G}_{(3)}$ is the mode-3 matricization of tensor $\mathcal{G}$ and matrix $\mathbf{D}$ is 3-D difference operator. The augmented Lagrangian function for (21) is

$$\begin{aligned}& L_A(\mathcal{X},\mathcal{A},\mathcal{B},\mathcal{F},\mathcal{Z},\mathcal{C},\mathbf{W},\mathbf{H},\mathbf{S}) \\ &= \frac{1}{2}\|\mathcal{Y}_h - \mathcal{A}\times_1 \mathbf{P_1}\times_2 \mathbf{P_2}\|_F^2 + \frac{\lambda_1}{2}\|\mathcal{Y}_m - \mathcal{B}\times_3 \mathbf{P_3}\|_F^2 \\ &\quad + \lambda_2\|\mathcal{F}\|_1 + \frac{\lambda_3}{2}\|\mathbf{G}_{(3)} - \mathbf{E}\mathbf{U_1}\|_F^2 + \lambda_4\|\mathbf{V_1}\|_{\text{MCP}} \\ &\quad + \frac{\lambda_5}{2}\|(\mathbf{Y}_h)_{(3)} - \mathbf{E}\mathbf{U_2}\|_F^2 + \lambda_6\|\mathbf{V_2}\|_{\text{MCP}} \\ &\quad + \frac{\beta}{2}\left\|\mathcal{A} - \mathcal{X} - \frac{\boldsymbol{\lambda}^{\mathcal{A}}}{\beta}\right\|^2 + \frac{\beta}{2}\left\|\mathcal{B} - \mathcal{X} - \frac{\boldsymbol{\lambda}^{\mathcal{B}}}{\beta}\right\|^2\end{aligned}$$

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$$+\frac{\beta}{2}\left\|\mathcal{F}-\mathbf{D}(\mathcal{Z})-\frac{\boldsymbol{\lambda}^{\mathcal{F}}}{\beta}\right\|^2+\frac{\beta}{2}\left\|\mathcal{Z}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{Z}}}{\beta}\right\|^2$$
$$+\frac{\beta}{2}\left\|\mathcal{G}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{G}}}{\beta}\right\|^2+\frac{\beta}{2}\left\|\mathbf{U_1}-\mathbf{V_1}-\frac{\boldsymbol{\lambda}^{\mathbf{U}_1}}{\beta}\right\|^2$$
$$+\frac{\beta}{2}\left\|\mathbf{U_2}-\mathbf{V_2}-\frac{\boldsymbol{\lambda}^{\mathbf{U}_2}}{\beta}\right\|^2$$

where $\boldsymbol{\lambda}^{\mathcal{A}}$, $\boldsymbol{\lambda}^{\mathcal{B}}$, $\boldsymbol{\lambda}^{\mathcal{F}}$, $\boldsymbol{\lambda}^{\mathcal{Z}}$, $\boldsymbol{\lambda}^{\mathcal{G}}$, $\boldsymbol{\lambda}^{\mathbf{U}_1}$, and $\boldsymbol{\lambda}^{\mathbf{U}_2}$ are the Lagrange multiplier vectors, and $\beta$ is the positive penalty scalar. Based on multiblock ADMM, we can approximately solve this optimization problem by alternatively minimizing one variable with the others fixed. The optimization problem of $L_A$ with respect to each variable can be solved by the following subproblems.

*1) $\mathcal{X}, \mathcal{C}_p, \mathbf{W}_p, \mathbf{H}_p, \mathbf{S}_p, \mathbf{N}_p$ Sub-Problem:* The subproblem to optimize $L_A$ with respect to $\mathcal{X}, \mathcal{C}_p, \mathbf{W}_p, \mathbf{H}_p, \mathbf{S}_p, \mathbf{N}_p$ is

$$\begin{aligned}\min_{\substack{\mathcal{X}_p,\mathcal{C}_p,\mathbf{W}_p,\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{N}_p}} &\frac{\beta}{2}\|\mathcal{A}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{A}}}{\beta}\|^2+\frac{\beta}{2}\|\mathcal{B}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{B}}}{\beta}\|^2\\ &+\frac{\beta}{2}\|\mathcal{Z}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{Z}}}{\beta}\|^2+\frac{\beta}{2}\|\mathcal{G}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{G}}}{\beta}\|^2\\ \text{s.t. } &\mathcal{X}=\mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1\mathbf{W}_p\times_2\mathbf{H}_p\times_3\mathbf{S}_p\times_4\mathbf{N}_p\}_{p=1,2,\ldots,K})\\ &\mathbf{W}_p^T\mathbf{W}_p=\mathbf{I},\quad \mathbf{H}_p^T\mathbf{H}_p=\mathbf{I},\ \mathbf{S}_p^T\mathbf{S}_p=\mathbf{I},\ \mathbf{N}_p^T\mathbf{N}_p=\mathbf{I}.\end{aligned}\tag{22}$$

The above equation can be easily transformed into the following equivalent problem:

$$\begin{aligned}\min_{\substack{\mathcal{C}_p,\mathbf{W}_p\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{V}_p}} &2\beta\|\mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1\mathbf{W}_p\times_2\mathbf{H}_p\times_3\mathbf{S}_p\\ &\times_4\mathbf{N}_p\}_{p=1,2,\ldots,N})-\mathcal{L}\|^2\\ \text{s.t. } &\mathbf{W}_p^T\mathbf{W}_p=\mathbf{I},\quad \mathbf{H}_p^T\mathbf{H}_p=\mathbf{I},\ \mathbf{S}_p^T\mathbf{S}_p=\mathbf{I},\ \mathbf{V}_p^T\mathbf{V}_p=\mathbf{I}\end{aligned}\tag{23}$$

where $\mathcal{L}=(1/4\beta)((\beta\mathcal{A}-\boldsymbol{\lambda}^{\mathcal{A}})+(\beta\mathcal{B}-\boldsymbol{\lambda}^{\mathcal{B}})+(\beta\mathcal{Z}-\boldsymbol{\lambda}^{\mathcal{Z}}))+(\beta\mathcal{G}-\boldsymbol{\lambda}^{\mathcal{G}})$. This problem can be rewritten as

$$\begin{aligned}\min_{\substack{\mathcal{C}_p,\mathbf{W}_p\\ \mathbf{H}_p,\mathbf{S}_p,\mathbf{V}_p}} &\sum_{p=1}^{K}\frac{1}{2}\|\mathcal{C}_p\times_1\mathbf{W}_p\times_2\mathbf{H}_p\times_3\mathbf{S}_p\times_4\mathbf{V}_p-\mathbf{Q}_p(\mathcal{L})\|^2\\ \text{s.t. } &\mathbf{W}_p^{\mathrm{T}}\mathbf{W}_p=\mathbf{I},\quad \mathbf{H}_p^{\mathrm{T}}\mathbf{H}_p=\mathbf{I},\ \mathbf{S}_p^{\mathrm{T}}\mathbf{S}_p=\mathbf{I},\ \mathbf{V}_p^{\mathrm{T}}\mathbf{V}_p=\mathbf{I}.\end{aligned}\tag{24}$$

Then similar to the Joint HOOI Algorithm in [52], this problem can be approximately solved by alternatively updating the following formulas:

$$\mathcal{C}_p=\mathbf{Q}_p(\mathcal{L})\times_1\mathbf{W}_p^T\times_2\mathbf{H}_p^T\times_3\mathbf{S}_p^T\times_4\mathbf{N}_p^T\tag{25}$$
$$\mathbf{W}_p=\mathrm{SVD}\big(\big(\mathbf{Q}_p(\mathcal{L})\times_2\mathbf{H}_p^T\times_3\mathbf{S}_p^T\times_4\mathbf{N}_p^T\big)_{(1)},r_1\big)\tag{26}$$
$$\mathbf{H}_p=\mathrm{SVD}\big(\big(\mathbf{Q}_p(\mathcal{L})\times_1\mathbf{W}_p^T\times_3\mathbf{S}_p^T\times_4\mathbf{N}_p^T\big)_{(2)},r_2\big)\tag{27}$$
$$\mathbf{S}_p=\mathrm{SVD}\big(\big(\mathbf{Q}_p(\mathcal{L})\times_1\mathbf{W}_p^T\times_2\mathbf{H}_p^T\times_4\mathbf{N}_p^T\big)_{(3)},r_3\big)\tag{28}$$
$$\mathbf{N}_p=\mathrm{SVD}\big(\big(\mathbf{Q}_p(\mathcal{L})\times_1\mathbf{W}_p^T\times_2\mathbf{H}_p^T\times_3\mathbf{S}_p^T\big)_{(4)},r_4\big)\tag{29}$$

where $\mathrm{SVD}(\mathbf{A},r)$ indicates top $r$ singular vectors of matrix $\mathbf{A}$. Then $\mathcal{X}$ can be updated by $\mathcal{X}=\mathcal{Q}^{-1}(\{\mathcal{C}_p\times_1\mathbf{W}_p\times_2\mathbf{H}_p\times_3\mathbf{S}_p\times_4\mathbf{N}_p\}_{p=1,2,\ldots,K})$.

*2) $\mathcal{A}$ Subproblem:* Optimizing $L_A$ with respect to $\mathcal{A}$ can be written as

$$\min_{\mathcal{A}}\frac{1}{2}\|\mathcal{Y}_h-\mathcal{A}\times_1\mathbf{P_1}\times_2\mathbf{P_2}\|_F^2+\frac{\beta}{2}\left\|\mathcal{A}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{A}}}{\beta}\right\|^2\tag{30}$$

which can be formalized as the following matrix format:

$$\min_{\mathbf{A}_{(3)}}\frac{1}{2}\|(\mathbf{Y}_h)_{(3)}-\mathbf{A}_{(3)}\mathbf{M}\|_F^2+\frac{\beta}{2}\left\|\mathbf{A}_{(3)}-\mathbf{X}_{(3)}-\frac{\boldsymbol{\lambda}_{(3)}^{\mathcal{A}}}{\beta}\right\|^2\tag{31}$$

where $\mathbf{A}_{(3)}, (\mathbf{Y}_h)_{(3)}, \boldsymbol{\lambda}_{(3)}^{\mathcal{A}}, \mathbf{X}_{(3)}$ are the three-mode unfolding matrices of $\mathcal{A}, \mathcal{Y}_h, \boldsymbol{\lambda}^{\mathcal{A}}, \mathcal{X}$, respectively, and $\mathbf{M}$ is the spatial downsampling matrix. The problem in (31) can be treated as solving the following linear system:

$$(\mathbf{M}\mathbf{M}^T+\beta\mathbf{I})\mathbf{A}_{(3)}^T=\beta\mathbf{X}_{(3)}^T+\big(\boldsymbol{\lambda}_{(3)}^{\mathcal{A}}\big)^T+\mathbf{M}((\mathbf{Y}_h)_{(3)})^T.\tag{32}$$

Obviously, this linear system can be solved by off-the-shelf conjugate gradient techniques.

*3) $\mathcal{B}$ Subproblem:* Optimizing $L_A$ with respect to $\mathcal{B}$ can be formalized as

$$\min_{\mathcal{B}}\frac{\lambda_1}{2}\|\mathcal{Y}_m-\mathcal{B}\times_3\mathbf{P_3}\|_F^2+\frac{\beta}{2}\left\|\mathcal{B}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{B}}}{\beta}\right\|^2\tag{33}$$

which can be matriculated as follows:

$$\min_{\mathbf{B}_{(3)}}\frac{\lambda_1}{2}\|(\mathbf{Y}_m)_{(3)}-\mathbf{P_3}\mathbf{B}_{(3)}\|_F^2+\frac{\beta}{2}\left\|\mathbf{B}_{(3)}-\mathbf{X}_{(3)}-\frac{\boldsymbol{\lambda}_{(3)}^{\mathcal{B}}}{\beta}\right\|^2$$

where $\mathbf{B}_{(3)}, (\mathbf{Y}_m)_{(3)}, \boldsymbol{\lambda}_{(3)}^{\mathcal{B}}, \mathbf{X}_{(3)}$ are the mode-3 unfolding matrices of $\mathcal{B}, \mathcal{Y}_m, \boldsymbol{\lambda}^{\mathcal{B}}, \mathcal{X}$, respectively, and $\mathbf{P_3}$ is the spectral downsampling matrix. The problem in (34) can be treated as solving the following linear system:

$$(\lambda_1\mathbf{P_3}^*\mathbf{P_3}+\beta\mathbf{I})\mathbf{B}_{(3)}=\beta\mathbf{X}_{(3)}+\boldsymbol{\lambda}_{(3)}^{\mathcal{B}}+\lambda_1\mathbf{P_3}^*(\mathbf{Y}_m)_{(3)}\tag{34}$$

where the superscript $*$ denotes the adjoint matrix. Obviously, this linear system can be solved by off-the-shelf conjugate gradient techniques.

*4) $\mathcal{Z}$ Subproblem:* The subproblem to optimize $L_A$ with respect to $\mathcal{Z}$ has the following form:

$$\min_{\mathcal{Z}}\frac{\beta}{2}\left\|\mathcal{F}-\mathbf{D}(\mathcal{Z})-\frac{\boldsymbol{\lambda}^{\mathcal{F}}}{\beta}\right\|^2+\frac{\beta}{2}\left\|\mathcal{Z}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{Z}}}{\beta}\right\|^2.\tag{35}$$

The problem in (35) can be treated as solving the following linear system:

$$(\beta\mathbf{D}^*\mathbf{D}+\beta\mathbf{I})\mathcal{Z}=\beta\mathcal{X}+\boldsymbol{\lambda}^{\mathcal{Z}}+\mathbf{D}^*(\beta\mathcal{F}-\boldsymbol{\lambda}^{\mathcal{F}}).\tag{36}$$

Thanks to the block-circulant structure of the matrix corresponding to the operator $D^*D$, it can be diagonalized by the 3-D FFT matrix. Therefore, $\mathcal{Z}$ can be fast computed by

$$\mathrm{ifftn}\left(\frac{\mathrm{fftn}(\beta\mathcal{X}+\boldsymbol{\lambda}^{\mathcal{Z}}+\mathbf{D}^*(\beta\mathcal{F}-\boldsymbol{\lambda}^{\mathcal{F}}))}{\beta\mathbf{1}+\beta(|\mathrm{fftn}(\mathbf{D}_h)|^2+|\mathrm{fftn}(\mathbf{D}_v)|^2+|\mathrm{fftn}(\mathbf{D}_t)|^2)}\right)\tag{37}$$

where fftn and ifftn, respectively, indicate fast 3-D Fourier transform and its inverse transform, $|\cdot|^2$ is the element-wise square, and the division is also performed element-wisely. Note that the denominator in the equation can be precalculated outside the main loop, avoiding the extra computational cost.

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*5) $\mathcal{F}$ Subproblem:* Optimizing $L_A$ with respect to $\mathcal{F}$ can be written as

$$\min_{\mathcal{F}} \frac{\beta}{2}\left\|\mathcal{F}-\mathbf{D}(\mathcal{Z})-\frac{\boldsymbol{\lambda}^{\mathcal{F}}}{\beta}\right\|^2+\lambda_2\|\mathcal{F}\|_1. \tag{38}$$

This subproblem can be solved by the well-known soft shrinkage operator as follows:

$$\mathcal{F}=\text{soft}\left(\mathbf{D}(\mathcal{Z})+\frac{\boldsymbol{\lambda}^{\mathcal{F}}}{\beta},\frac{\lambda_2}{\beta}\right) \tag{39}$$

where $\text{soft}(\mathbf{a},\tau):=\text{sgn}(\mathbf{a})\cdot\max(|\mathbf{a}|-\tau,0)$.

*6) $\mathcal{G}$ Subproblem:* The subproblem to optimize $L_A$ with respect to $\mathcal{G}$ is

$$\min_{\mathcal{G}} \frac{\lambda_3}{2}\|\mathbf{G}_{(3)}-\mathbf{E}\mathbf{U_1}\|^2+\frac{\beta}{2}\left\|\mathcal{G}-\mathcal{X}-\frac{\boldsymbol{\lambda}^{\mathcal{G}}}{\beta}\right\|^2 \tag{40}$$

which is equivalent to

$$\min_{\mathcal{G}}\left\|\mathcal{G}-\frac{1}{\lambda_3+\beta}\left(\lambda_3\text{fold}_3(\mathbf{E}\mathbf{U_1})+\beta\left(\mathcal{X}+\frac{\boldsymbol{\lambda}^{\mathcal{G}}}{\beta^{\mathcal{G}}}\right)\right)\right\|^2. \tag{41}$$

This problem has the following closed-form solution:

$$\mathcal{G}=\frac{1}{\lambda_3+\beta}\left(\lambda_3\text{fold}_3(\mathbf{E}\mathbf{U_1})+\beta\left(\mathcal{X}+\frac{\boldsymbol{\lambda}^{\mathcal{G}}}{\beta}\right)\right). \tag{42}$$

*7) $\mathbf{U}_1,\mathbf{U}_2$ Subproblem:* Optimizing $L_A$ with respect to $\mathbf{U_1}$ can be formalized as

$$\min_{\mathbf{U_1}} \frac{\lambda_3}{2}\|\mathbf{G}_{(3)}-\mathbf{E}\mathbf{U_1}\|^2+\frac{\beta}{2}\left\|\mathbf{U_1}-\mathbf{V_1}-\frac{\boldsymbol{\lambda}^{\mathbf{U_1}}}{\beta}\right\|^2. \tag{43}$$

This problem can be treated as solving the following linear system:

$$(\lambda_3\mathbf{E}^T\mathbf{E}+\beta\mathbf{I})\mathbf{U_1}=\beta\mathbf{V_1}+\boldsymbol{\lambda}^{\mathbf{U_1}}+\lambda_3\mathbf{E}^T\mathbf{G}_{(3)} \tag{44}$$

where $\mathbf{I}$ is the identity matrix. This linear system can be solved by off-the-shelf conjugate gradient techniques.

The subproblem to optimize $L_A$ with respect to $\mathbf{U_2}$ has a similar form

$$\min_{\mathbf{U_2}} \frac{\lambda_5}{2}\|(\mathbf{Y}_h)_{(3)}-\mathbf{E}\mathbf{U_2}\|^2+\frac{\beta}{2}\left\|\mathbf{U_2}-\mathbf{V_2}-\frac{\boldsymbol{\lambda}^{\mathbf{U_2}}}{\beta}\right\|^2. \tag{45}$$

This problem can also be treated as solving a linear system

$$(\lambda_2\mathbf{E}^T\mathbf{E}+\beta\mathbf{I})\mathbf{U_2}=\beta\mathbf{V_2}+\boldsymbol{\lambda}^{\mathbf{U_2}}+\lambda_5\mathbf{E}^T(\mathbf{Y}_h)_{(3)}. \tag{46}$$

*8) $\mathbf{V}_1,\mathbf{V}_2$ Subproblem:* The subproblem to optimize $L_A$ with respect to $\mathbf{V_1}$ is

$$\begin{aligned}&\min_{\mathbf{V_1}} \lambda_4\|\mathbf{V_1}\|_{\text{MCP}}+\frac{\beta}{2}\left\|\mathbf{U_1}-\mathbf{V_1}-\frac{\boldsymbol{\lambda}^{\mathbf{U_1}}}{\beta}\right\|^2\\&\text{s.t. } \mathbf{V_1}\geq 0.\end{aligned} \tag{47}$$

This problem can be solved using the iteratively reweighted $\ell_1$ norm method as follows:

$$\begin{aligned}\mathbf{V_1}&=\left[\text{soft}\left(\mathbf{U_1}-\frac{\boldsymbol{\lambda}^{\mathbf{U_1}}}{\beta},\frac{\lambda_4}{\beta}\right)\right]_+\\\boldsymbol{\Omega}&=\psi'_{a,\lambda}(\mathbf{V_1})\\\mathbf{V_1}&=\left[\text{soft}\left(\mathbf{U_1}-\frac{\boldsymbol{\lambda}^{\mathbf{U_1}}}{\beta},\frac{\lambda_4}{\beta}\boldsymbol{\Omega}\right)\right]_+\end{aligned} \tag{48}$$

TABLE I

QUANTITATIVE RESULTS OF THE TESTED METHODS ON THE PAVIA UNIVERSITY, PAVIA CENTRE, AND SALINAS DATA SETS

| Methods | Pavia University | | | | | | |
|---|---|---|---|---|---|---|---|
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| CNMF | 2.997 | 38.598 | 0.952 | 1.969 | 1.072 | 0.970 | 2.208 |
| HSRCSU | 3.192 | 38.050 | 0.958 | 1.774 | 1.155 | 0.967 | 2.164 |
| NSSR | 2.669 | 39.603 | 0.966 | 1.741 | 0.943 | 0.977 | 1.905 |
| CSTF | 2.549 | 40.003 | 0.961 | 1.687 | 0.921 | 0.976 | 1.899 |
| OURS | **2.114** | **41.627** | **0.978** | **1.298** | **0.756** | **0.985** | **1.454** |
| Methods | Pavia Centre | | | | | | |
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| CNMF | 2.853 | 39.025 | 0.964 | 2.063 | 1.162 | 0.978 | 2.084 |
| HSRCSU | 2.985 | 38.633 | 0.973 | 1.807 | 1.229 | 0.978 | 1.913 |
| NSSR | 2.452 | 40.340 | 0.977 | 1.874 | 0.995 | 0.984 | 1.783 |
| CSTF | 2.230 | 41.164 | 0.975 | 1.748 | 0.909 | 0.985 | 1.682 |
| OURS | **2.039** | **41.943** | **0.982** | **1.369** | **0.827** | **0.989** | **1.451** |
| Methods | Salinas | | | | | | |
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| CNMF | 3.312 | 37.729 | 0.941 | 1.669 | 0.933 | 0.953 | 2.115 |
| HSRCSU | 3.825 | 36.478 | 0.948 | 1.638 | 1.182 | 0.952 | 2.186 |
| NSSR | 2.935 | 38.776 | 0.954 | 1.498 | 0.823 | 0.963 | 1.770 |
| CSTF | 2.926 | 38.805 | 0.952 | 1.488 | 0.815 | 0.963 | 1.768 |
| OURS | **2.775** | **39.266** | **0.963** | **1.399** | **0.795** | **0.970** | **1.523** |

where $\psi'_{a,\lambda}(\cdot)$ is the derivative of MCP penalty function and computed element-wisely for the matrix $\mathbf{V_1}$, $\text{soft}(\cdot,\cdot)$ is the soft shrinkage operator as in (39), and $[x]_+=\max\{x,0\}$.

Similarly, optimizing $L_A$ with respect to $\mathbf{V_2}$ can be written as

$$\begin{aligned}&\min_{\mathbf{V_2}} \lambda_6\|\mathbf{V_2}\|_{\text{MCP}}+\frac{\beta}{2}\left\|\mathbf{U_2}-\mathbf{V_2}-\frac{\boldsymbol{\lambda}^{\mathbf{U_2}}}{\beta}\right\|^2\\&\text{s.t. } \mathbf{V_2}\geq 0.\end{aligned} \tag{49}$$

This problem can also be solved as in (48)

$$\begin{aligned}\mathbf{V_2}&=\left[\text{soft}\left(\mathbf{U_2}-\frac{\boldsymbol{\lambda}^{\mathbf{U_2}}}{\beta},\frac{\lambda_6}{\beta}\right)\right]_+\\\boldsymbol{\Omega}&=\psi'_{a,\lambda}(\mathbf{V_2})\\\mathbf{V_2}&=\left[\text{soft}\left(\mathbf{U_2}-\frac{\boldsymbol{\lambda}^{\mathbf{U_2}}}{\beta},\frac{\lambda_6}{\beta}\boldsymbol{\Omega}\right)\right]_+.\end{aligned} \tag{50}$$

*9) $\mathbf{E}$ Subproblem:* Optimizing $L_A$ with respect to $\mathbf{E}$ can be written as

$$\begin{aligned}&\min_{\mathbf{E}} \frac{\lambda_3}{2}\|\mathbf{G}_{(3)}-\mathbf{E}\mathbf{U_1}\|^2+\frac{\lambda_5}{2}\|(\mathbf{Y}_h)_{(3)}-\mathbf{E}\mathbf{U_2}\|^2\\&\text{s.t. } \mathbf{E}\geq 0\end{aligned} \tag{51}$$

which is equivalent to the following optimization problem:

$$\begin{aligned}&\mathbf{E}=\arg\min_{\mathbf{E}} \|\mathbf{G}_{(3)}-\mathbf{E}\mathbf{U_1}\|^2+\frac{\lambda_5}{\lambda_3}\|(\mathbf{Y}_h)_{(3)}-\mathbf{E}\mathbf{U_2}\|^2\\&\text{s.t. } \mathbf{e}_k\geq 0\end{aligned} \tag{52}$$

where $\mathbf{e}_k$ is the $k$th column of $\mathbf{E}$. To reduce the computational complexity, similar to ODL method [70] and the spectral dictionary learning method in [34], instead of updating the whole $\mathbf{E}$ at once, we can solve (52) by using block coordinate descent [71], that is, during each iteration we update only one column of $\mathbf{E}$ while keeping the others fixed under the nonnegative constraint. Let $\mathbf{E}^{(t)}$ denotes the dictionary obtained after the $t$th iteration, $\boldsymbol{\alpha}_k$ denotes the $k$th row of $\mathbf{U_1}$, $\boldsymbol{\beta}_k$ denotes the $k$th row of $\mathbf{U_2}$, and let $\mathbf{e}_k^{(t+1)}=\mathbf{e}_k^{(t)}+\Delta\mathbf{e}_k$. Then $\Delta\mathbf{e}_k$ can be

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TABLE II

QUANTITATIVE RESULTS OF THE SIMPLIFIED AND ORIGINAL VERSION OF OUR METHOD ON THE PAVIA UNIVERSITY, PAVIA CENTRE, AND SALINAS DATA SETS

| Methods | Pavia University | | | | | | |
|---|---|---|---|---|---|---|---|
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| Simplified Version | 2.133 | 41.550 | **0.978** | 1.307 | 0.759 | **0.985** | 1.457 |
| Original Version | **2.114** | **41.627** | **0.978** | **1.298** | **0.756** | **0.985** | **1.454** |
| Methods | Pavia Centre | | | | | | |
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| Simplified Version | 2.148 | 41.491 | 0.980 | 1.424 | 0.858 | 0.988 | 1.481 |
| Original Version | **2.039** | **41.943** | **0.982** | **1.369** | **0.827** | **0.989** | **1.451** |
| Methods | Salinas | | | | | | |
| | RMSE | PSNR | SSIM | SAM | ERGAS | UIQI | DD |
| Simplified Version | 2.834 | 39.083 | 0.960 | 1.408 | 0.798 | 0.967 | 1.574 |
| Original Version | **2.775** | **39.266** | **0.963** | **1.399** | **0.795** | **0.970** | **1.523** |

TABLE III

PARAMETERS AND THEIR CORRESPONDING VALUES IN ALGORITHM 1

| Trade-off | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\lambda_4$ | $\lambda_5$ | $\lambda_6$ |
|---|---|---|---|---|---|---|
| | 1 | 0.0001 | 0.2 | 0.0002 | 0.1 | 0.0001 |
| 3D patch | $w_{patch}$ | $S_{patch}$ | $N_{patch}$ | $d_{patch}$ | | |
| | 8 | 36 | 45 | 7 | | |
| Redundancy | $r_1$ | $r_2$ | $r_3$ | $r_4$ | | |
| | 8 | 8 | 3 | 21 | | |

obtained by solving the following optimization problem:

$$\begin{aligned}\Delta\mathbf{e}_k = \arg\min_{\Delta\mathbf{e}_k} &\|\mathbf{G}_{(3)} - \mathbf{E}^{(t)}\mathbf{U_1} - \Delta\mathbf{e}_k\boldsymbol{\alpha}_k\|^2 \\ &+ \frac{\lambda_5}{\lambda_3}\|(\mathbf{Y}_h)_{(3)} - \mathbf{E}^{(t)}\mathbf{U_2} - \Delta\mathbf{e}_k\boldsymbol{\beta}_k\|^2 \\ \text{s.t. } &\mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0. \end{aligned} \tag{53}$$

Let $\mathbf{H_1}^{(t)} = \mathbf{G}_{(3)} - \mathbf{E}^{(t)}\mathbf{U_1}$ and $\mathbf{H_2}^{(t)} = (\mathbf{Y}_h)_{(3)} - \mathbf{E}^{(t)}\mathbf{U_2}$ denote the residual matrices after the $t$th iteration, then $\mathbf{H_1}^{(t)} \in \mathbf{R}^{S\times(W\times H)}$, $\mathbf{H_1}^{(t)} \in \mathbf{R}^{S\times(w\times h)}$. Let $n_1 = W \times H$, $n_2 = w \times h$, then problem (53) can be rewritten as follows:

$$\begin{aligned}&\Delta\mathbf{e}_k \\ &= \arg\min_{\Delta\mathbf{e}_k} \|\mathbf{H_1}^{(t)} - \Delta\mathbf{e}_k\boldsymbol{\alpha}_k\|^2 + \frac{\lambda_5}{\lambda_3}\|(\mathbf{H_2}^{(t)} - \Delta\mathbf{e}_k\boldsymbol{\beta}_k\|^2 \\ &\quad \text{s.t. } \mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0 \\ &= \arg\min_{\Delta\mathbf{e}_k} \sum_{s=1}^{S}\Bigg(\sum_{i=1}^{n_1}\left(\left(\mathbf{H_1}_{s,i}^{(t)}\right) - \Delta\mathbf{e}_k(s)\boldsymbol{\alpha}_k(i)\right)^2 \\ &\qquad + \frac{\lambda_5}{\lambda_3}\sum_{j=1}^{n_2}\left(\left(\mathbf{H_2}_{s,j}^{(t)}\right) - \Delta\mathbf{e}_k(s)\boldsymbol{\beta}_k(j)\right)^2\Bigg) \\ &\quad \text{s.t. } \mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0 \\ &= \arg\min_{\Delta\mathbf{e}_k} \sum_{s=1}^{S}\Bigg(\sum_{i=1}^{n_1}\boldsymbol{\alpha}_k(i)^2\Delta\mathbf{e}_k(s)^2 + \frac{\lambda_5}{\lambda_3}\sum_{j=1}^{n_2}\boldsymbol{\beta}_k(j)^2\Delta\mathbf{e}_k(s)^2 \\ &\qquad - 2\left(\boldsymbol{\alpha}_k\left(\mathbf{H_1}_{s,:}^{(t)}\right)^T + \frac{\lambda_5}{\lambda_3}\boldsymbol{\beta}_k\left(\mathbf{H_2}_{s,:}^{(t)}\right)^T\right)\Delta\mathbf{e}_k(s)\Bigg) \\ &\qquad + \text{Constant} \\ &\quad \text{s.t. } \mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0 \\ &= \arg\min_{\Delta\mathbf{e}_k} \sum_{s=1}^{S}(\Delta\mathbf{e}_k(s)) - \frac{\boldsymbol{\alpha}_k\left(\mathbf{H_1}_{s,:}^{(t)}\right)^T + \frac{\lambda_5}{\lambda_3}\boldsymbol{\beta}_k\left(\mathbf{H_2}_{s,:}^{(t)}\right)^T}{\sum_{i=1}^{n_1}\boldsymbol{\alpha}_k(i)^2 + \frac{\lambda_5}{\lambda_3}\sum_{j=1}^{n_2}\boldsymbol{\beta}_k(j)^2})^2 \\ &\qquad + \text{Constant} \\ &\quad \text{s.t. } \mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0 \\ &= \arg\min_{\Delta\mathbf{e}_k} \left\|\Delta\mathbf{e}_k - \frac{\mathbf{H_1}^{(t)}\boldsymbol{\alpha}_k{}^T + \frac{\lambda_5}{\lambda_3}\mathbf{H_2}^{(t)}\boldsymbol{\beta}_k{}^T}{\sum_{i=1}^{n_1}\boldsymbol{\alpha}_k(i)^2 + \frac{\lambda_5}{\lambda_3}\sum_{j=1}^{n_2}\boldsymbol{\beta}_k(j)^2}\right\|_2^2 \\ &\qquad + \text{Constant} \\ &\quad \text{s.t. } \mathbf{e}_k^{(t)} + \Delta\mathbf{e}_k \geq 0 \end{aligned} \tag{54}$$

**Algorithm 1** Optimization Algorithm for the Model (20)

**Input:** LR-HS image $\mathcal{Y}_h$; HR-MS image $\mathcal{Y}_m$; Spatial downsampling matrix $\mathbf{M}$; Spectral downsampling matrix $\mathbf{P_3}$; Algorithm parameters $\lambda_1$, $\lambda_2$, $\lambda_3$, $\lambda_4$, $\lambda_5$, $\lambda_6$, $r_1$, $r_2$, $r_3$, $r_4$.

**Initialization:** $\mathcal{X}$, $\mathbf{U_1}$, $\mathbf{U_2}$ and $\mathbf{E}$ are randomly initialized; $\mathcal{A}$, $\mathcal{B}$, $\mathcal{Z}$, $\mathcal{G}$ are all initialized to be equal to $\mathcal{X}$; $\mathbf{V_1}$ and $\mathbf{V_2}$ are initialized to be equal to $\mathbf{U_1}$ and $\mathbf{U_2}$, respectively; $\mathcal{F}$ is initialized by 0.

**while** not converge **do**

1. Updating $\mathcal{X}$ and $\mathcal{C}_p$, $\mathbf{W}_p$, $\mathbf{H}_p$, $\mathbf{S}_p$, $\mathbf{N}_p$ by the following algorithm:

**while** not converge **do**

(1). Updating $\mathcal{C}_p$ via Eq.(25);

(2). Updating $\mathbf{W}_p$, $\mathbf{H}_p$, $\mathbf{S}_p$ and $\mathbf{V}_p$ via Eqs. (26), (27), (28) and (29).

**end while**

2. Updating $\mathcal{A}$ via Eq.(32);
3. Updating $\mathcal{B}$ via Eq.(34);
4. Updating $\mathcal{Z}$ via Eq.(37);
5. Updating $\mathcal{F}$ via Eq.(39);
6. Updating $\mathcal{G}$ via Eq.(42);
7. Updating $\mathbf{U_1}$, $\mathbf{U_2}$ via Eq.(44), (46), respectively;
8. Updating $\mathbf{V_1}$, $\mathbf{V_2}$ via Eqs.(48), (50), respectively;
9. Updating $\mathbf{E}$ via Eq.(55);
10. Updating multipliers via Eqs.(56).

**end while**

**Output:** The reconstructed HR-HS image $\mathcal{X}$.

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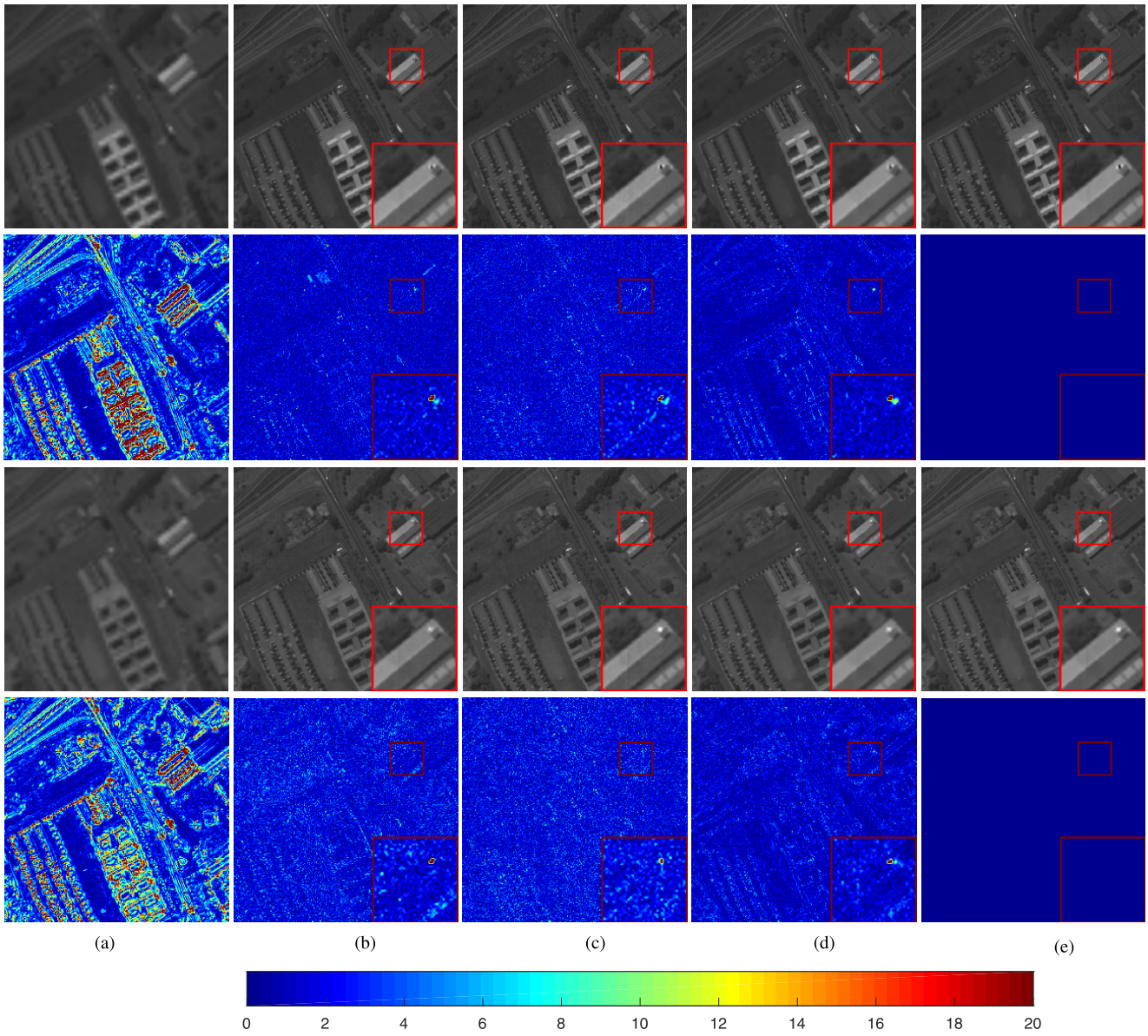



Fig. 1. Reconstructed images and corresponding error images of Pavia University for the 20th and 60th bands. The first and second rows show the reconstructed images for the 20th band and corresponding error images, respectively; The third and fourth rows show the reconstructed images for the 60th band and corresponding error images, respectively. (a) LR-HS image. (b) NSSR method. (c) CSTF method. (d) Our method. (e) Ground truth.

where $\mathbf{H_1}_{s,i}^{(t)}$ and $\mathbf{H_2}_{s,j}^{(t)}$ denote the $(s,i)$th element of $\mathbf{H_1}^{(t)}$ and the $(s,j)$th element of $\mathbf{H_2}^{(t)}$, respectively; $\boldsymbol{\alpha}_k(i)$, $\boldsymbol{\beta}_k(j)$ and $\Delta\mathbf{e}_k(s)$denote the $i$th element of $\boldsymbol{\alpha}_k$, the $j$th element of $\boldsymbol{\beta}_k$ and the $s$th element of $\Delta\mathbf{e}_k$, respectively; $\mathbf{H_1}_{s,:}^{(t)}$ and $\mathbf{H_2}_{s,:}^{(t)}$ denote the $s$th row of $\mathbf{H_1}^{(t)}$ and the $s$th row of $\mathbf{H_2}^{(t)}$, respectively. Based on (54), we can see that the solution of (53) can be given by

$$\mathbf{e}_k^{(t+1)} = \left[\mathbf{e}_k^{(t)} + \frac{\mathbf{H_1}^{(t)}\boldsymbol{\alpha}_k{}^T + \frac{\lambda_5}{\lambda_3}\mathbf{H_2}^{(t)}\boldsymbol{\beta}_k{}^T}{\sum_{i=1}^{n_1}\boldsymbol{\alpha}_k(i)^2 + \frac{\lambda_5}{\lambda_3}\sum_{j=1}^{n_2}\boldsymbol{\beta}_k(j)^2}\right]_+. \quad (55)$$

*10) Updating Multipliers:* According to the ADMM, the multipliers associated with $L_A$ are updated by the following formulas:

$$\begin{cases}\boldsymbol{\lambda}^{\mathcal{A}} \leftarrow \boldsymbol{\lambda}^{\mathcal{A}} - \gamma\beta(\mathcal{A}-\mathcal{X}) \\ \boldsymbol{\lambda}^{\mathcal{B}} \leftarrow \boldsymbol{\lambda}^{\mathcal{B}} - \gamma\beta(\mathcal{B}-\mathcal{X}) \\ \boldsymbol{\lambda}^{\mathcal{Z}} \leftarrow \boldsymbol{\lambda}^{\mathcal{Z}} - \gamma\beta(\mathcal{Z}-\mathcal{X}) \\ \boldsymbol{\lambda}^{\mathcal{F}} \leftarrow \boldsymbol{\lambda}^{\mathcal{F}} - \gamma\beta(\mathcal{F}-\mathbf{D}(\mathcal{Z})) \\ \boldsymbol{\lambda}^{\mathcal{G}} \leftarrow \boldsymbol{\lambda}^{\mathcal{G}} - \gamma\beta(\mathcal{G}-\mathcal{X}) \\ \boldsymbol{\lambda}^{\mathbf{U}_1} \leftarrow \boldsymbol{\lambda}^{\mathbf{U}_1} - \gamma\beta(\mathbf{U_1}-\mathbf{V_1}) \\ \boldsymbol{\lambda}^{\mathbf{U}_2} \leftarrow \boldsymbol{\lambda}^{\mathbf{U}_2} - \gamma\beta(\mathbf{U_2}-\mathbf{V_2})\end{cases} \quad (56)$$

where $\gamma$ is a parameter associated with the convergence rate of the algorithm. The proposed algorithm for model (20) can now be summarized in Algorithm 1.

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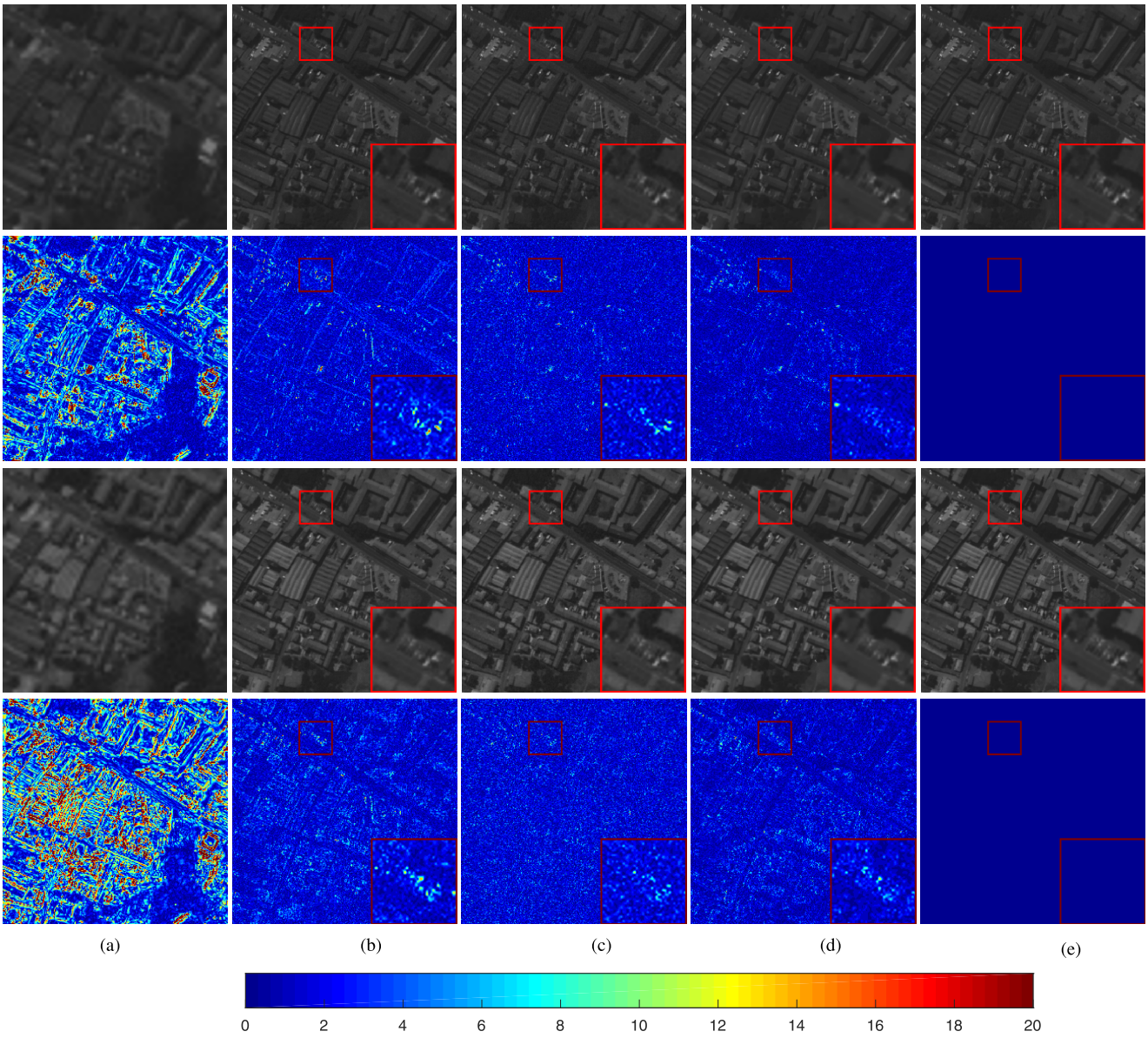



Fig. 2. Reconstructed images and corresponding error images of Pavia Centre for the 10th and 50th bands. The first and second rows show the reconstructed images for the 10th band and corresponding error images, respectively. The third and fourth rows show the reconstructed images for the 50th band and corresponding error images, respectively. (a) LR-HS image. (b) NSSR method. (c) CSTF method. (d) Our method. (e) Ground truth.

### *B. Computational Cost Analysis*

For getting the HR-HSI with a size of $W \times H \times S$, the main per-iteration cost of Algorithm 1 lies in the subproblems 1), 2), 3), 4), and 7), since other subproblems can be solved through a simple algebraic calculation. As for subproblem 1), the main cost lies in the joint HOOI algorithm, which requires performing several SVDs for each $d \times d \times H \times S$ tensor $\mathcal{Q}_p(\mathcal{X})$. Thus, if we adopt a parallel computing procedure, the cost of solving this subproblem is comparable to a simple HOOI algorithm, which gives the cost as $O(T(d^2r_2r_3r_4 + (r_2r_3r_4)^3 + d^2r_1r_3r_4 + (r_1r_3r_4)^3 + H^2r_1r_2r_4 + (r_1r_2r_4)^3 + S^2r_1r_2r_3 + (r_1r_2r_3)^3))$, where $T$ is the number of iterations used in HOOI. As for subproblems 2), 3), and 7), the main cost lies in solving a large linear system by using conjugate gradient technique. It is known that the computational cost of conjugate gradient method for solving a linear system $Ar = b$ is $O(m\sqrt{k})$, where $m$ is the number of nonzero entries in $A$ and $k$ is its condition number. Thus, the main cost solving such three subproblems is $O((WH)^2\sqrt{k_1} + S^2(\sqrt{k_2} + \sqrt{k_3} + \sqrt{k_4}))$, where $k_1$, $k_2$, $k_3$ and $k_4$ are the condition numbers for the linear systems (32), (34), (44), and (46), respectively. It is not hard to see that these condition numbers could not be very large, since all such linear systems are well defined. As for subproblems 4), the main cost lies in performing FFT. Considering the fact that the computational complexity of FFT is $O(N \log N)$, where $N$ is the data size. As such, the cost of solving subproblem 4) is $O(WHS \log(WHS))$. In all, the computational complexity of Algorithm 1 is $O(T(d^2r_2r_3r_4 + (r_2r_3r_4)^3 + d^2r_1r_3r_4 + (r_1r_3r_4)^3 + H^2r_1r_2r_4 + (r_1r_2r_4)^3 + S^2r_1r_2r_3 + (r_1r_2r_3)^3) + (WH)^2\sqrt{k_1} + S^2(\sqrt{k_2} + \sqrt{k_3} + \sqrt{k_4}) + WHS \log(WHS))$, which is comparable with the existing tensor-based method for dealing with

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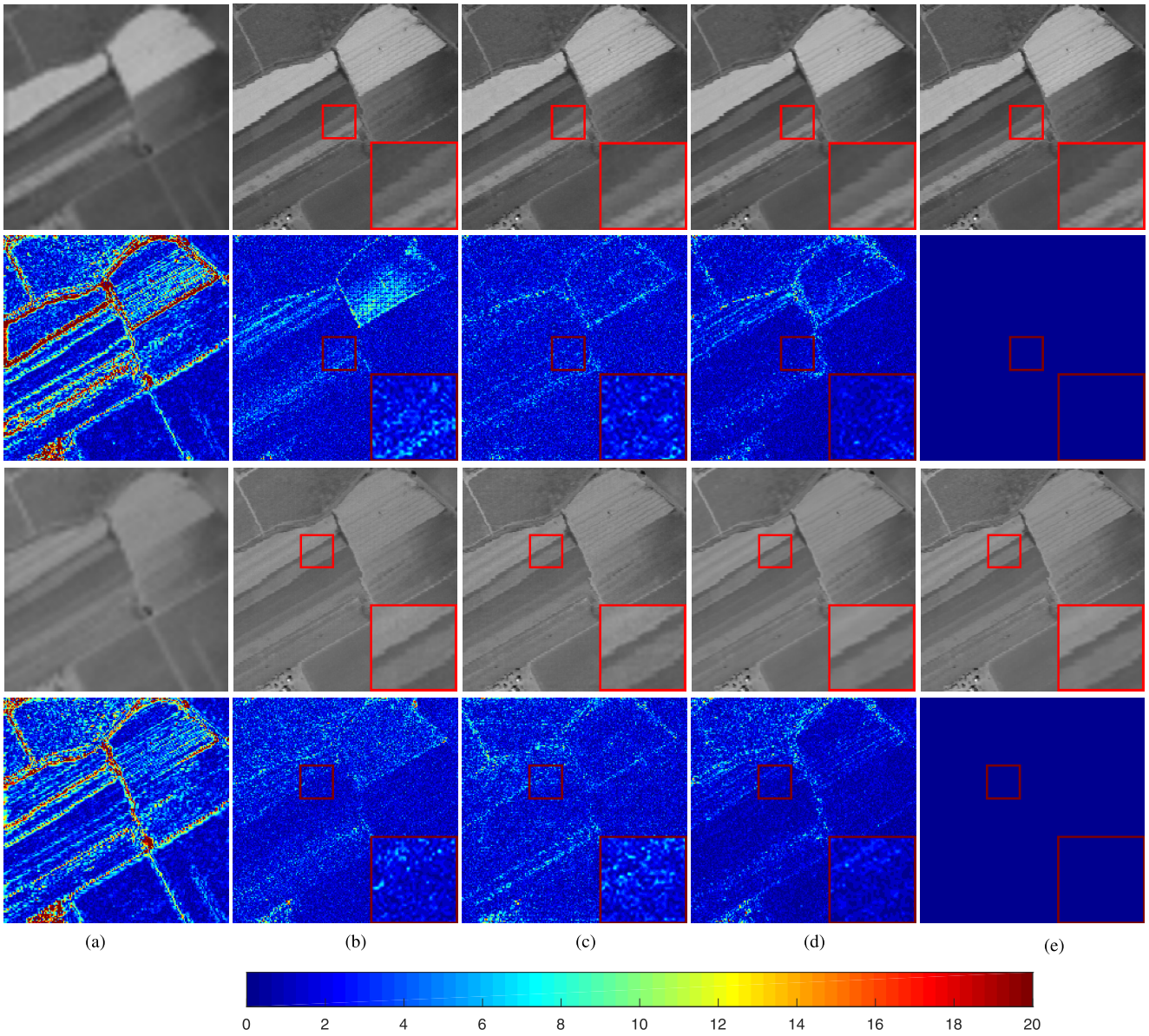



Fig. 3. Reconstructed images and corresponding error images of Salinas for the 30th and 80th bands. The first and second rows show the reconstructed images for the 30th band and corresponding error images, respectively; The third and fourth rows show the reconstructed images for the 80th band and corresponding error images, respectively. (a) LR-HS image. (b) NSSR method. (c) CSTF method. (d) Our method. (e) Ground truth.

other HSI tasks, see [45], [46], and thus reasonable in practice.

## V. Experiments

### A. Compared Methods and Performance Evaluation Measures

In this section, the performances of our proposed algorithm are compared with several state-of-the-art HS and MS image fusion methods, including the CNMF method [29], the HS coupled spectral unmixing (HSRCSU) method [32], the NSSR method [34], and the CSTF method [38]. The assessments used to evaluate the quality of reconstructed HS images contain the root mean square error (RMSE), the peak SNR (PSNR), structural similarity index (SSIM), spectral angle mapper (SAM) [72], relative dimensionless global error in synthesis (ERGAS) [73], universal image quality index (UIQI) [74], the degree of distortion (DD), and the correlation coefficient (CC). It is known that the higher values of PSNR, SSIM, UIQI, and CC and the smaller values of RMSE, SAM, ERGAS, and DD show better performances.

### B. Experimental Data Sets

Four different data sets of HS images are used to compare the performances of our proposed method and other fusion methods. The first data set is the Pavia University image [75] acquired by the reflective opticssystem imaging spectrometer (ROSIS) optical sensor over the urban area of the University of Pavia, Italy. This image is of size $610 \times 340 \times 115$ with a spatial resolution of 1.3 m, and the number of bands has been reduced to 93 bands after removing the water vapor absorption

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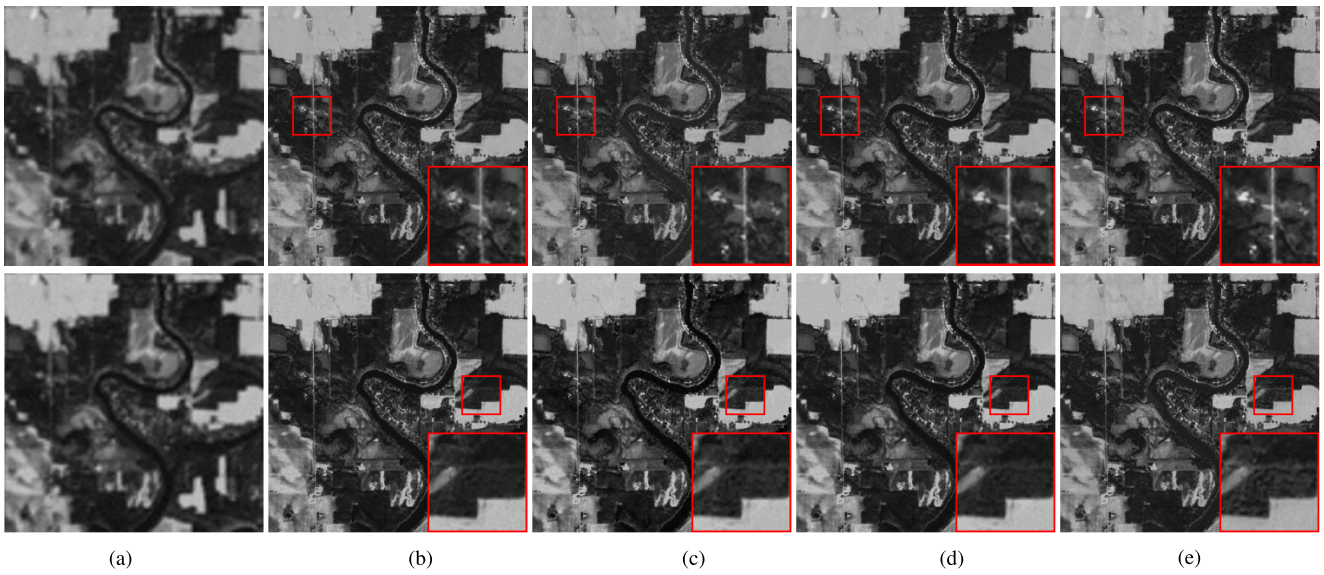



Fig. 4. Reconstructed images of Hyperion data set for the first and 20th bands. The first row shows the reconstructed images for the first band and the second row shows the reconstructed images for the 20th band. (a) LR-HS image. (b) CNMF method. (c) HSRCSU method. (d) NSSR method. (e) Our method.

bands (with spectral range from 0.43 to 0.86 $\mu$m). A subimage with the size of 256 $\times$ 256 $\times$ 93 is used as the groundtruth HR-HS image. Then the corresponding LR-HS image of size 64 $\times$ 64$\times$ 93 is constructed by applying a 5 $\times$ 5 Gaussian spatial filter with standard deviation 2.5 on each band of the HR-HS image and downsampling every four pixels in both height and width directions. A four-band HR-MS image of size 256 $\times$ 256 $\times$ 4 is constructed by filtering the HR-HS image with a IKONOS like reflectance spectral responses [27]. And following [72], the LR-HS and HR-MS image are both contaminated by zero-mean additive Gaussian noises, where SNR = 35 dB for the first 43 bands and SNR = 30 dB for the remaining 50 bands of the LR-HS image while SNR = 30 dB for all bands of the HR-MS image.

The second data set is the Pavia Centre image which is also acquired by the ROSIS sensor during a flight campaign over Pavia, Italy. After removing the water vapor absorption bands and some of the samples containing no information, Pavia Centre is of size 1096 $\times$ 715$\times$ 92. A subimage with a size of 256 $\times$ 256 $\times$ 92 is used as the ground truth HR-HS image, and the corresponding contaminated LR-HS and HR-MS image are constructed as in the Pavia University image.

The third data set is the Salinas image which is collected by the NASA's Airborne Visible and Infrared Imaging Spectrometer (AVIRIS) [76] over Salinas Valley, California. This image is of size 512$\times$217$\times$224 covering the wavelength range 400 nm-2500 nm with 10 nm sampling interval. The number of bands is reduced to 194 bands after removing the bands covering heavy noise and water absorption, then a subimage with the size of 200 $\times$ 200 $\times$ 194 is used as the ground-truth HR-HS image. The corresponding LR-HS image of size 50$\times$50$\times$194 is constructed by applying a 5$\times$5 Gaussian kernel with a standard deviation of 2.5 followed by a downsampling factor of 4. The HR-MS image is generated by downsampling the original images in the spectral domain using the spectral transform operator of the Landsat TM [29], where the spectral response is transformed according to Landsat TM bands 1–5 and 7, covering the spectral regions of 450–520, 520–600, 630–690, 760–900, 1550–1750, and 2080–2350 nm [77]. Then the LR-HS and HR-MS image are both contaminated by zero-mean additive Gaussian noises, where SNR = 40 dB for the LR-HS image and SNR = 35 dB for the HR-MS image.

The fourth data set is a real data set for HS and MS image fusion problem. The HS image is taken by the Hyperion sensor loaded on Earth Observing-1 satellite, which is of size 2350 $\times$ 990 $\times$ 220 with a spatial resolution of 30 m, and the number of bands has been reduced to 89 bands after removing the noisy bands. A subimage with a size of 100 $\times$ 100 $\times$ 89 is used as the LR-HS image. The MS image with 13 bands is collected by the Sentinel-2A satellite, and among those bands we use four ones with spatial resolution of 10 m. The spatial downsampling factor is 3 and thus the size of HR-MS image is 300 $\times$ 300 $\times$ 4.

### C. Experimental Results

In this section, we shall compare the fusion results of the five tested methods using the aforementioned data sets.

Table I shows the RMSE, PSNR, SSIM, SAM, ERGAS, UIQI, DD, and CC of the reconstructed HR-HS image by CNMF, HSRCSU, NSSR, CSTF, and our method for the Pavia University data set, where the best results are marked in bold for clarity. As we can see from Table I, our method can get the best results for all the assessment indices in this data set, which shows the superiority of our method to the other fusion methods in terms of reconstruction accuracy. This is because we have fully exploited the underlying priors of the targeted HR-HS image, including spatial–spectral correlation, nonlocal self-similarity, and spatial–spectral continuity. In addition, the term of spectral unmixing based on MCP regularization has

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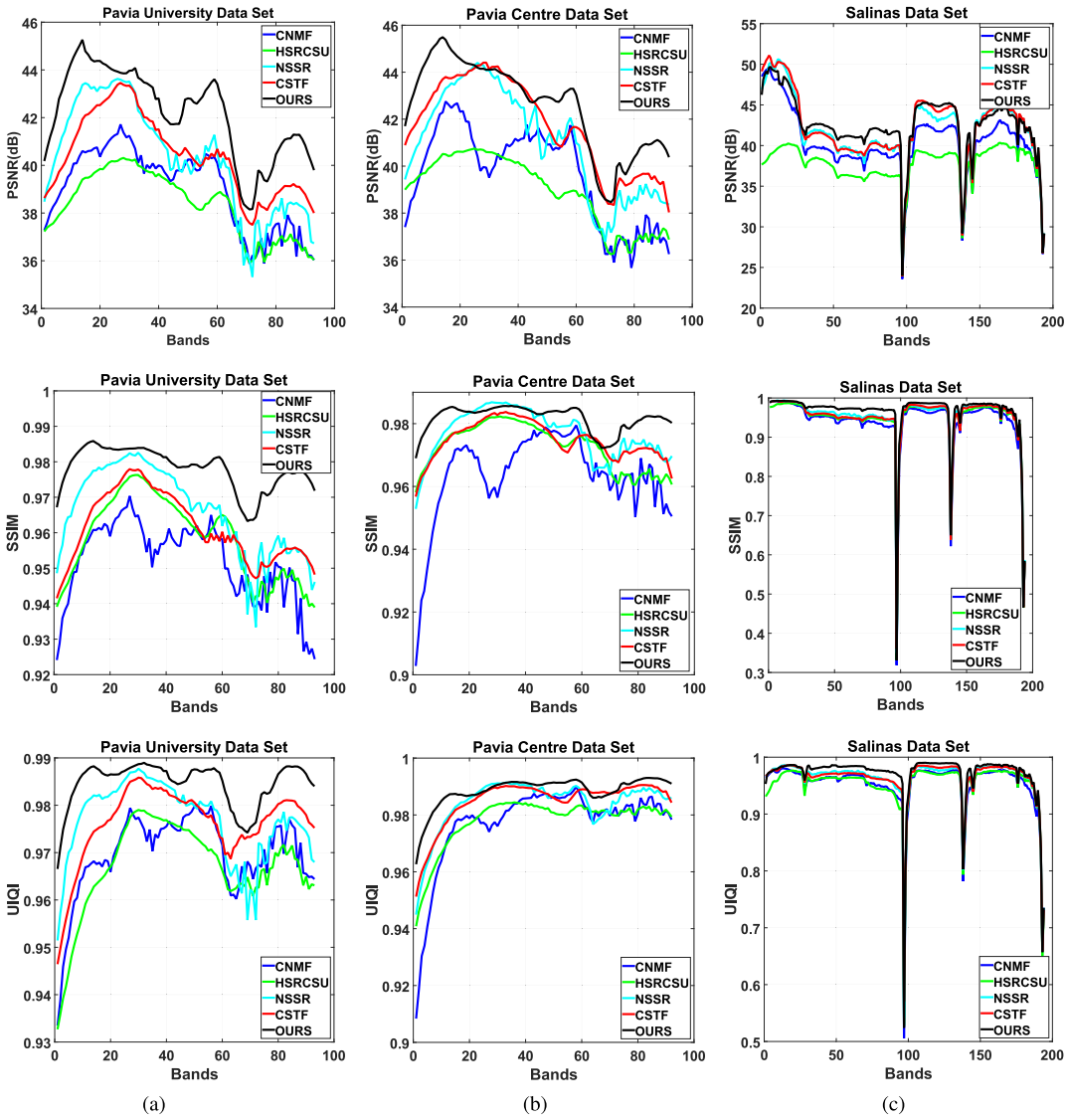



Fig. 5. PSNR, SSIM, and UIQI values of each band in (a) Pavia University, (b) Pavia Centre, and (c) Salinas data sets.

effectively reduced the spectral distortions to further improve the reconstruction performances.

In order to further intuitively compare the performances of those fusion methods, we show the 20th and 60th bands and the corresponding error images reconstructed by NSSR, CSTF, and our method in Fig. 1. From Fig. 1, we can see while all these tested methods can well reconstruct the spatial structures of the HR-HS image, our method obviously outperforms the others in recovering the details of the original HR-HS image.

Table I shows the quantitative results of the Pavia Centre data set. We can see from Table I that our method gets the best scores for all the assessments. The 10th and 50th bands and the corresponding error images reconstructed by NSSR, CSTF, and our method are shown in Fig. 2, in which we can see that our method outperforms the other two methods in recovering image details.

The quantitative results of those tested methods on the Salinas data set are also shown in Table I. The corresponding error images for the 30th and 80th bands reconstructed by NSSR, CSTF, and our method are shown in Fig. 3. We can see that our method can get the best reconstruction results both in terms of quantitative assessments and recovering image details.

To further compare the performances of those fusion methods for the real data set, we show the first and 20th bands fused by CNMF, HSRCSU, NSSR, and our method in Fig. 4. In this case, the spatial and spectral responses in the degradation model are unknown, and we estimate them from the observed data pair as done in [78] and [79]. Specifically, separability assumption for the spatial response is necessary for the CSTF method and unfortunately, the estimated one is not separable, thus we do not consider CSTF for comparison in this case. One can see from Fig. 4 that our method can achieve better image details, which also demonstrate its effectiveness for real fusion problems.

Fig. 5(a)–(c) shows the PSNR, SSIM, and UIQI curves as functions of the spectral bands over the Pavia University, Pavia Centre, and Salinas data set for the test methods. It can be

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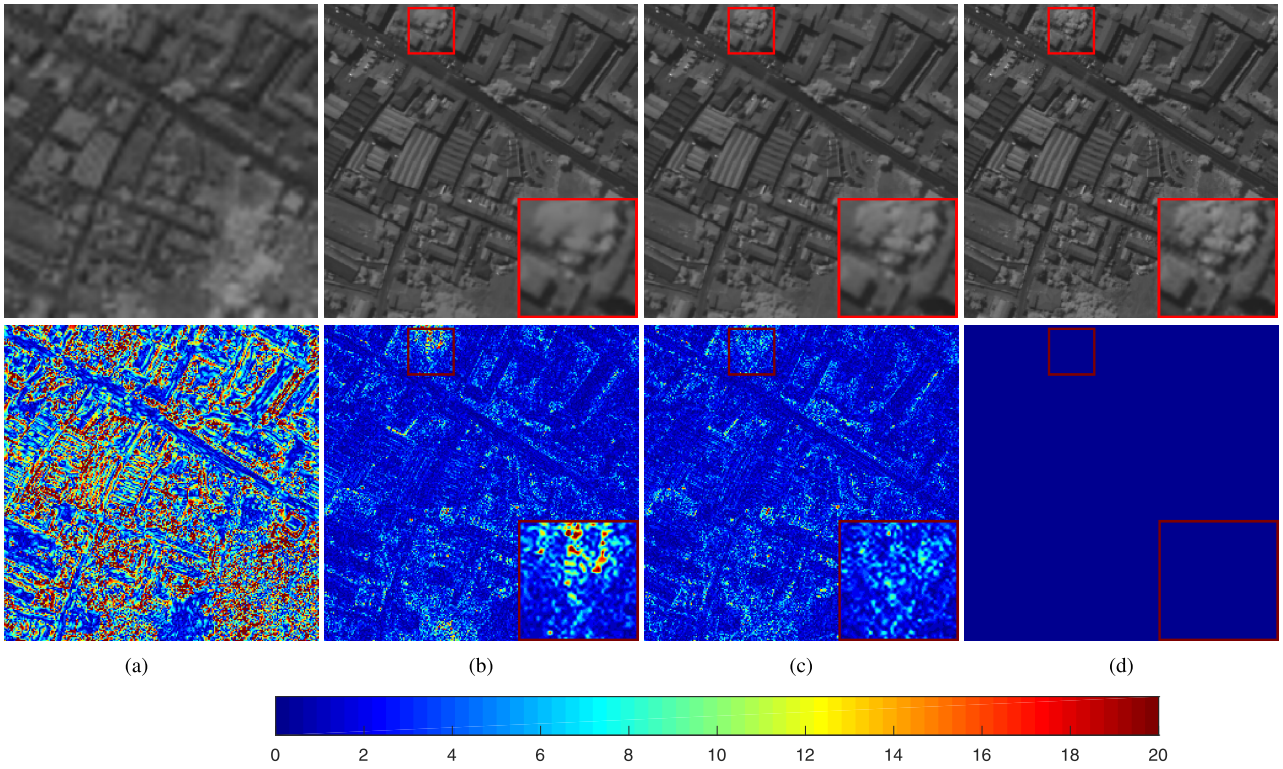



Fig. 6. Reconstructed images and corresponding error images of Pavia Centre for the 70th band. (a) LR-HS image. (b) Simplified version of our method. (c) Original version of our method. (d) Ground truth.

seen that our method consistently outperforms other methods at most of the spectral bands for all the three data sets.

To verify the effectiveness of the spectral unmixing term in our model, we further compare our method with a simplified version of our model without the spectral unmixing term, that is, let the parameters $\lambda_3 = \lambda_4 = \lambda_5 = \lambda_6 = 0$ in our model (20), and the quantitative results are shown in Table II. We can see from Table II that the spectral unmixing term in our model is assuredly able to improve the performances of our method. And as an example, Fig. 6 shows the 70th bands of Pavia Centre data set and corresponding error images reconstructed by the original version and simplified version (without the spectral unmixing term) of our method. We can see that compared with the simplified version, the original version method performs better in recovering the details in the image, which further proves the effectiveness of the spectral unmixing term in our model.

### *D. Empirical Analysis for Algorithm Convergence*

We provide an empirical analysis for the convergence of the proposed optimization algorithm on the three test data sets. We use the relative change relChgX $:= ((\|\mathcal{X}^k - \mathcal{X}^{k-1}\|_F)/\|\mathcal{X}^{k-1}\|_F)$, the relative error relErrX $:= ((\|\mathcal{X}^k - \mathcal{X}_0\|_F)/\|\mathcal{X}_0\|_F)$, the objective function value in model (20) and the PSNR value as the four assessment indices of algorithm convergence, where $\mathcal{X}^k$ is the recovered result in $k$th iteration and $\mathcal{X}_0$ is the ground truth HR-HS image.

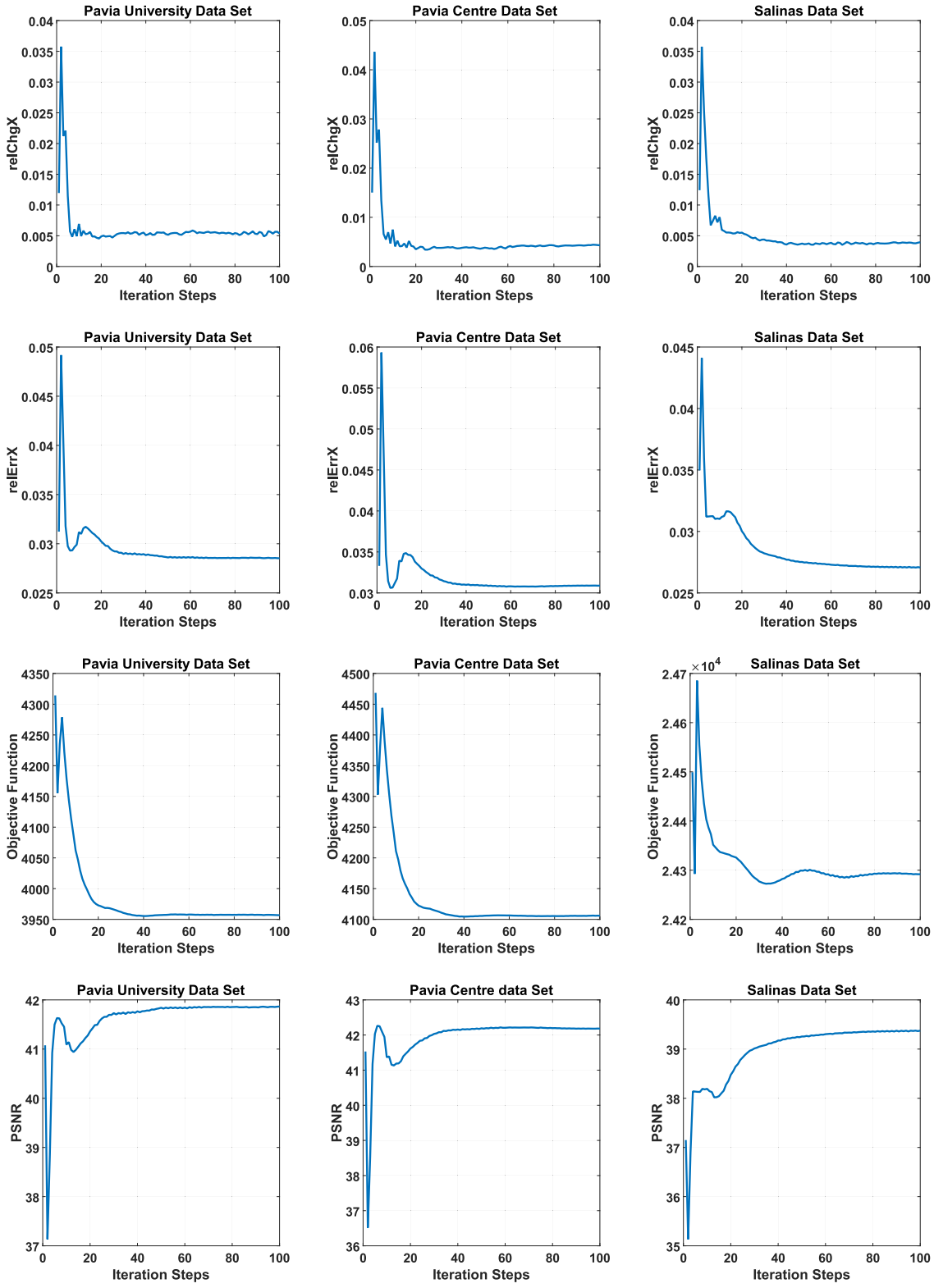
In Fig. 7, we show the curves of the relative change, the relative error, the objective function value, and the PSNR as functions of the iteration steps in the proposed optimization algorithm for the three test data sets. We observe from Fig. 7 that the curves of all assessment indexes come to a stable value when the algorithm reach a relatively high iteration number, which suggests that the proposed algorithm converges well empirically.

### *E. Implementation Issues and Parameters Discussions*

In Algorithm 1, those parameters can be divided into three groups based on their characteristics: 1) $\lambda_1$, $\lambda_2$, $\lambda_3$, $\lambda_4$, $\lambda_5$ and $\lambda_6$ provide a tradeoff between the reconstruction error term, the 3D-TV term and the spectral unmixing term in the model, in our experiments those parameters are empirically fixed, that is, $\lambda_1 = 1$, $\lambda_2 = 10^{-4}$, $\lambda_3 = 0.2$, $\lambda_4 = 0.001 * \lambda_3$, $\lambda_5 = 0.1$, $\lambda_6 = 0.001 * \lambda_5$; 2) the size of 3-D patch $d$, the size of search window around one patch $S_{\text{patch}}$, the number of collected similar 3-D patches $N_{\text{patch}}$, and the sliding distance $w_{\text{patch}}$ are the parameters to construct the nonlocal low rank 4-D tensors, where $d$, $S_{\text{patch}}$, $N_{\text{patch}}$, and $w_{\text{patch}}$ are empirically set as 8, 36, 45, and 7, respectively, [80], [81]; and 3) $r_1$, $r_2$, $r_3$, $r_4$ control the complexity of spatial redundancy, spectral redundancy and similarity of 3-D patches, respectively, where $r_1$, $r_2$, and $r_4$ are empirically fixed as 8, 8, ceil[1] $(N_{\text{patch}} * 0.45)$, respectively. Actually, $r_3$ is the only parameter that needs to tune carefully in our method. $r_3$ controls the spectral correlation of the reconstructed HR-HS image and it may vary for different HS images obviously. We empirically found that $r_3$ can be tuned

[1]ceil($a$) indicates the smallest integer larger than $a$.

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(a) Pavia University (b) Pavia Centre (c) Salinas

Fig. 7. Empirical analysis of algorithm convergence. The four rows represent the relChgX, relErrX, objection function values, and PSNR curves, respectively, as functions of the iteration steps in the proposed optimization algorithm in (a) Pavia University, (b) Pavia Centre, and (c) Salinas data sets.

in the range [3,10] for various HS images, and for the three test data sets in our experiments $r_3 = 3$ is a proper choice. All the parameters in Algorithm 1 and their corresponding values in our experiments are summarized in Table III. Besides, to initialize the $\mathbf{U}_1$, $\mathbf{U}_2$, and $\mathbf{E}$ in Algorithm 1, the number of endmembers $J$ should be determined first. In all our

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experiments, we do not tune $J$ and fix it as 120. It is worth noting that in order to reduce the computational cost, the clustering is performed by the $K$-nearest neighbor method. Specifically, for each 3-D patch of the patch set $\mathcal{G}$, we search $N_{\text{patch}}$ similar 3-D patches as a cluster from a big window around this 3-D patch.

## VI. Conclusion

This article presents an effective HS and MS image fusion method based on nonlocal low-rank tensor decomposition and spectral unmixing. To fully exploit the high spatial–spectral correlations and the nonlocal self-similarity of the HR-HS image, we first segment it into many overlapped 3-D patches and cluster those patches into some clusters, which can be considered as four-order tensors. Then the Tucker decomposition is used to formulate the low-rank property of those tensors, and the corresponding MS image is used as a data consistency constraint to maintain the spatial information. Meanwhile, a 3D-TV is used as a regularizer to exploit the spatial–spectral continuity of the HS image. Finally, an MCP regularization-based linear spectral unmixing is applied as a significant spectral regularizer to reduce spectral distortions. Based on the aforementioned prior knowledge, a unified optimization model is constructed to formulate the HS and MS fusion problem and an efficient ADMM-based algorithm is developed to solve the resulting model. Experimental results on three different data sets demonstrate that the proposed method can achieve smaller reconstruction errors and better visual quality than several state-of-the-art HS and MS fusion methods.

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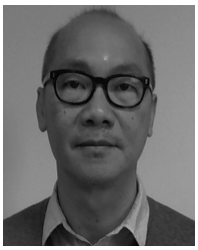
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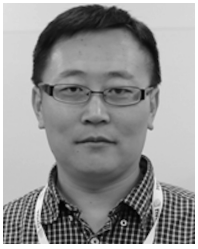
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